

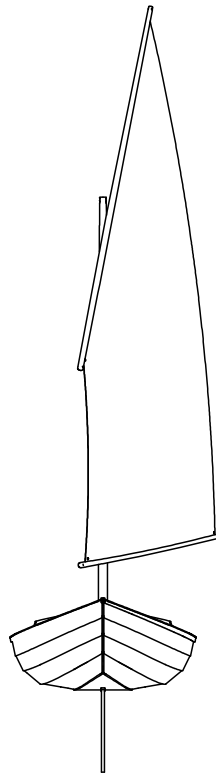
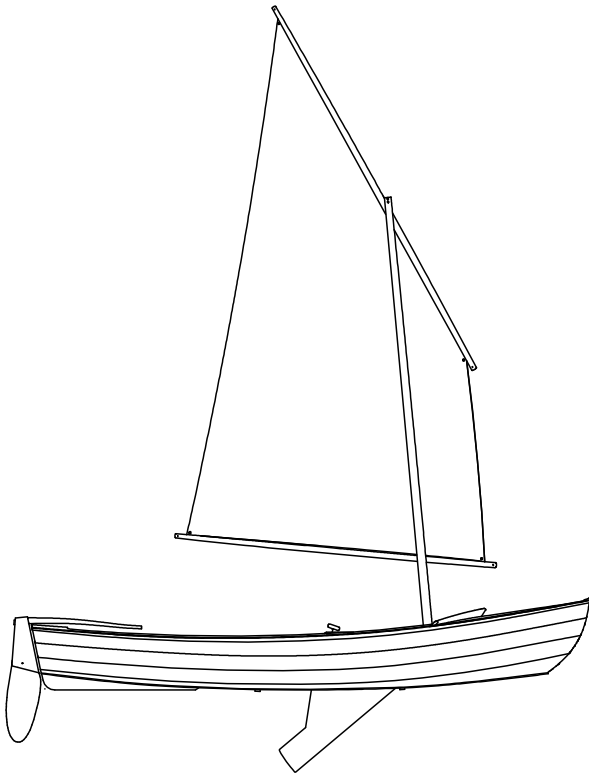
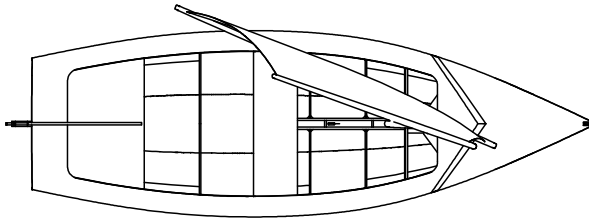
Phoenix 15

LOA - 4.637m (15' 2-1/2")

BOA - 1.547m (5' 1")

Draft - 135mm (5-5/16")

Displacement at DWL - 345kg (759lbs) in salt water



designed by Ross Lillistone

Phoenix 15 – A boat for Sail, Oars and Small Outboard



"What do you mean by a beach cruiser?" inquired Briggs.

"It means a boat to cruise along the beaches in shallow water," Goddard replied, "a boat to sleep aboard when hauled out on the beach, and I can tell you that this is an interesting and risky sort of cruising. It takes skill and experience to sail close to the shore if it is a rock region and there is a sea running, but you can visit many unfrequented places in a beach cruiser. Of course, there are sheltered waters in the rivers and marshes where there is no danger...."

(The Compleat Cruiser by L. Francis Herreshoff – Sheridan House 1956 Library of Congress Catalog Card Number: 56-12511)

When I first read the words quoted above back in 1980 I had already been beach cruising for fifteen years, even though I didn't know that was what it was called. Being fortunate enough to have been brought-up on the shores of Moreton Bay in southern Queensland in a family interested in wooden small-craft I was blessed from the beginning.

At the time of my first reading of the wonderful book from which I quote, I was spending my spare time day-sailing and beach cruising the shores near Townsville in tropical North Queensland, around places such as Magnetic Island. The conditions in the region provided plenty of adventure and some danger due to the underwater inhabitants and the robust sea conditions.

However, my initial beach cruising adventures were modest, and carried out in tin canoes, in very small sailing boats such as the wonderful Charles G. MacGregor-designed *Sabot*, and a variety of old flat-bottomed rowing and outboard boats. As fate would have it, in 1974 I inherited a 15foot 2-1/2inch x 6ft cruising dinghy designed and built by my late father, Victor Ross Lillistone. She was conceived specifically for cruising and day-sailing, constructed from marine plywood over Western Red Cedar framing and longitudinal stringers. The narrow flat-bottom was from 1/2 inch ply and the four batten-seam planks each side were from 1/4 inch ply. The glue was Resorcinol and the fastenings silicon-bronze. She is still sailing today, and is being used by the fourth generation of our family. Her name is '**Phoenix**'.



Phoenix in earlier days

The very same *Phoenix* as she is today

I've done some pretty adventurous things in my life, but beach-cruising and day-sailing in simple dinghies has remained one of my greatest sources of satisfaction, and at age seventy, I intend to continue the activity for as long as I am physically able. All of this can be done by just about anybody, in a boat which they have built for themselves, at a small fraction of the cost of an off-the-shelf production boat.



Phoenix III

By about the year 2000, in the endless search for the 'Perfect Boat', I settled on the proportions that I felt were just right for beach cruising – 15ft x 5ft x 6ins (4.5m x 1.5m x 150mm). Enough length for a good displacement speed, fine entry lines to slice through a steep chop, narrow enough for light weight and efficient rowing with oars of a modest length, and shallow enough to be able to manoeuvre the boat over a reef or sand bank. A lightly built boat with these proportions would be able to carry three adults plus gear on protected waters, be effective as a beach-cruiser and day-sailer with two, and still be capable of being moved easily on the beach and sailed effectively by a solo skipper. On top of all that she should have provision for a realistic sleeping spot.



In early 2003 I was approached by a customer who has since become a long-term friend. His name is Paul Hernes, and he became the builder of the prototype of my beach-cruising/day-sailing design named '**Phoenix III**'. Since then, more than 300 sets of plans (and counting) have been purchased, with the design having been built world-wide.

Phoenix III sailing under the power of her sloop rig with a spritsail main



Phoenix III is constructed using the Glued-Lapstrake (i.e. Clinker) method, resulting in a hull which has a high strength-to-weight ratio, and even more importantly, a high stiffness-to-weight ratio. The epoxy-glued hull seams remain watertight for life, and for such a complex shape, the boat is relatively simple to build – forgiving work for an amateur who reads the extensive instruction manual, and speedy work for a professional.

Despite the quick and relatively simple construction, a building frame or 'strongback' must be made, with moulds, bulkheads and frames, over which the hull is assembled. Also, the shapes of the individual planks must be established using a procedure known as 'spiling'. None of these steps are difficult for someone who reads the instructions or one of the many books available, but it seems that potential builders can be intimidated by the process.

Over the years I've been approached by a number of people who have asked whether a CNC-cut (Computer Numerically Controlled) kit of parts is available and until recently the answer has been, "No". I have produced hand-cut kits for several of my other designs, but not for lap-strake (clinker) hulls.

But here is where fate stepped in once more... In recent years I have been in collaboration with Pat Reece, the owner of **Duck Flat Wooden Boats** in Adelaide. Pat is firmly focussed on quality design and construction, accompanied by a broadening of the Duck Flat Wooden Boats product range and service offerings. My involvement has been on the design and consultation side.

As part of this vigorous approach to being the 'One-Stop-Shop' for wooden boat enthusiasts, **Duck Flat Wooden Boats** is in the process of developing a comprehensive range of CNC-cut kits for amateur builders, with the latest version of the Phoenix range, **Phoenix 15** being the first new kit to reach the workshop floor. I am very proud to have been part of this ambitious project and I have enthusiastically worked on altering Phoenix III to incorporate what I have

learned since the first **Phoenix III** was drawn in 2003. Also, I have incorporated construction developments to make best use of CNC cutting and construction techniques.



Phoenix 15

Phoenix 15 kits are available in a range of modules from printed plans for home building through to complete boats ready to sail away. In-between those two extremes will be kits with the parts CNC-cut, including the lap-strake/clinker planks, all frames, bulkheads, stem components, transom, transom doublers, decks, seats and tank-tops, rigging components, quality fasteners, epoxy, glass cloth, rudder hardware, sails and more.

If you would like someone to do the hull construction for you, Duck Flat Wooden Boats will make a professionally built bare hull which you can finish at home. Best of all, the kits will be available in modular component packs so that you don't have to purchase an entire kit at the beginning if that doesn't suit you. All the way through the construction, assistance and advice will be available from the Duck Flat Wooden Boats team.

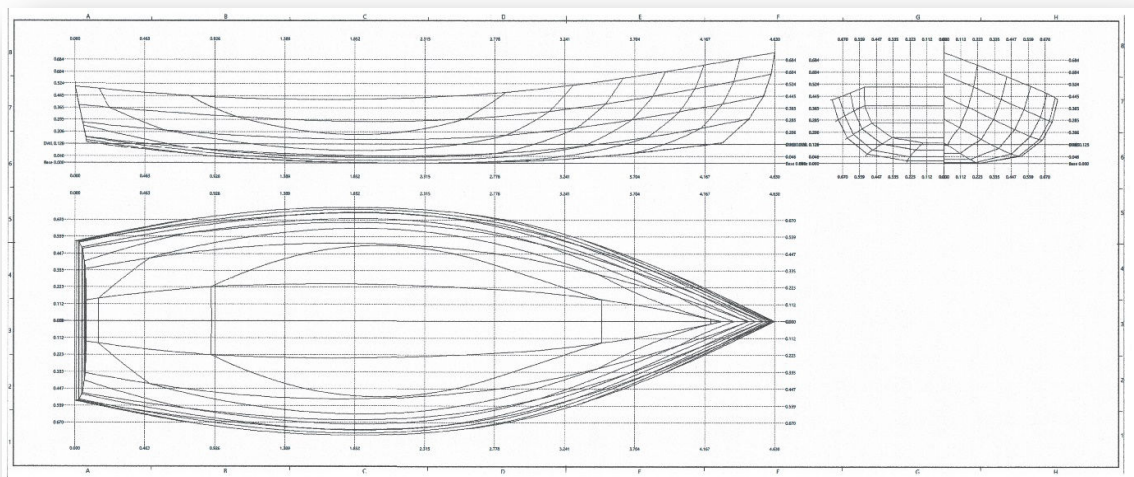
The differences between **Phoenix III** and the new **Phoenix 15** are minor in terms of hull shape, proportion and size. I would expect that no-one would be able to tell them apart if

two were floating side-by-side, and performance should be just about identical. In the case of **Phoenix 15**, the keelson and two garboard strakes (i.e. the lowest two planks abutting the keel) have been replaced by a narrow flat bottom panel of 12mm marine plywood. This change removes the two most difficult parts of construction – bevelling the keelson into the stem and fitting the garboard strakes. In addition, the resulting narrow bottom panel allows for easier fitting of the centreboard case, frames, and bulkheads, and for a substantially stronger bottom.



Phoenix 15

Other changes include a slight increase in breadth (beam) to 5ft (1.524m), a length between perpendiculars of 5ft 2ins (4.630m), and the addition of three very light frames. Although these alterations were introduced to make the construction better suited to CNC-cutting, I believe the boat has been improved slightly compared with her sister.



Phoenix 15 lines drawing

No matter whether you have a **Phoenix III** or a new **Phoenix 15**, the boat is a versatile, capable and pleasant vessel in which to enjoy our watery world with all the accompanying beaches, headlands, rivers and creeks. She will carry any of six different rigs, five of which use the very same mast, allowing an enthusiast to mix-and-match rigs. Better still, four of the current rig options do not require any standing rigging, so no need for stays and shrouds. This saves plenty of money in the first instance and makes for super-quick rigging and un-rigging.

The matter of rig choice is something which could take up a whole book, so I am not going to get into detail here. However, I will say that in the days when most sailing was done by people engaged in a struggle to make a living (fishermen, smugglers, ship's pilots, ferrymen, pirates and so on), rigs needed to be cheap to build, able to be maintained by their owners, reliable, versatile and easy to handle with the smallest crews possible. Racing has altered the evolutionary path of sailing boat rigs, and the need to gain every last second in the world of competition has seen enormous increases in the cost and complexity of sailing gear.

Cruising and recreational day-sailing is less about saving two seconds on the up-wind leg at all costs and more about elegant simplicity, reliability, lack of clutter, ease of maintenance, speed of setting and striking the rig, versatility, independence from shore-based repair organisations, and low cost. There is also great satisfaction to be gained from using your initiative to tweak rigs and experiment. The boat I started beach cruising in during very early seventies is now on her seventh different rig and we are still making experimental changes. With **Phoenix 15** all of these delights are at your fingertips.

Phoenix III tacking away from
the opposition

In this day and age most of us are subject to time pressure when it comes to recreational activity, and so auxiliary power is an important consideration when setting out for a day on the water. From the very start of my adventures in small boat design, I have paid close attention to rowing geometry. Not too many people seem to be interested in rowing these days, but that can be blamed on the disgraceful quality of the rowing equipment and oars which one can buy from chandleries and marine retailers. In addition, most rowing seems to be done in boats with hulls designed to run at planning speeds under the urge of outboard motors. Attempting to row a tinny or an



inflatable is discouraging enough but combining it with the use of badly shaped and unbalance oars ensures that the experience of rowing is unpleasant.

Phoenix 15 has a hull shape which is very well suited to rowing – so much so that leaving the rig at home and rowing for the pleasure of it is something I do regularly. The relationship between the location of the oarlocks and the rowing thwart must be designed properly, the standard of oarlocks and sockets should be high, and most important of all, oars should be correctly proportioned and well made. If all those elements are in place, you will find a boat such as **Phoenix 15** will allow you to row fast and far. True, it will take time to develop skill, but the results will pay-off for the rest of your life!



Phoenix III and Phoenix 15 have both been designed to be a pleasure to use under oars

(photo Tom Pamperin)

Despite my enthusiasm about sailing and rowing, auxiliary outboard power is sometimes a necessity for a variety of reasons which everybody can understand. **Phoenix 15** is equipped with an outboard motor mount on the transom, including a self-draining splash-well mounted to port of the rudder. This particular arrangement has been thoroughly tested in half-a-dozen or so of my designs and is much less obtrusive than the disgusting-looking metal outboard brackets which are so common.

Phoenix 15 has a slippery hull specifically designed for displacement speeds, and as such she needs very little power to run efficiently. For the last twenty-one years I've been using a four-stroke Honda 2hp (the current model is rated at 2.3hp) which weighs only 12.5kg (27-1/2lbs). I usually run at about one-third throttle at which speed the sound of the four-stroke is quite soothing and the boat runs at displacement speed (about 5.2knots or 9.6kph). Light weight is my primary consideration when it comes to an auxiliary outboard.



Under the power of a 2hp Honda outboard at part throttle

More recently I have been conducting tests to determine the practicality of electric auxiliary power. Electric trolling motors have proven to be unsuitable due to the large diameter fine-pitch propellers used for their primary function. However, for five or six years now I have been using a **Torqeedo Travel 1103CS** which is designed primarily as a propulsion motor rather than a trolling motor. The propeller technology comes from modern cargo ship design, and the integral lithium battery provides me with all the power I need. Intelligent operation is necessary because if you try to push past the displacement speed of the boat the electricity consumption increases exponentially, but at my boat/motor combination's sweet spot of about 4.2knots/7.8kph, I get around six hours endurance from a single battery which weighs 6kg/13.2lbs. Use a solar panel and the range increases substantially.

On the subject of the Torqeedo, I have been experimenting with e-motor-sailing, and the results have been spectacular. I've motor-sailed with internal combustion outboards, but the

noise is always there, and as the wind picks up the motor increases in revs and then decreases in revs as the wind moderates or you encounter some chop.

With the **Torqeedo Travel 1103CS** the situation is different. On calm or fluky wind days I set out with the rig up and the electric motor set to an input wattage of about 55 watts. This gives me a speed in a flat calm of about 2.7knots/5kph with total silence from the electric motor. As the wind increases the motor rpm stays almost constant (and silent) but the electricity consumption drops in inverse proportion to the wind speed. So, the boat gets along as though under sail power alone and you can get through the frustrating calms. Once the wind gets to the strength where the motor is consuming zero watts, you can lift to motor at your convenience. On a recent outing where I was e-motor-sailing for about three and a half hours with two people and gear aboard, I still had 90% battery capacity when I returned and had been able to cover a substantial distance instead of being held back by the calms. A perfect motor for your **Phoenix 15** – a motor which is also IP67 water-proof!

General Instructions

(Note: To illustrate processes, the photos used in this manual show a number of different boats, not just *Phoenix 15*)

Plywood

All plywood must be Marine Grade, stamped to comply with either BS1088 or AS/NZ2272. Do not compromise on the quality of your plywood, despite what the onlookers may tell you. Only ply made to the above mentioned standards is good enough for use in a boat. This has less to do with the type of glue used in the ply than with the standards required regarding approved species of timber, allowable faults in the veneers, number of veneers, thickness of veneers and so on. Once again, only BS1088 and AS/NZ2272 are good enough.

People from countries other than Australia and New Zealand need to research the national standards applicable to their location (I believe the US standard is APA 1-83). Whichever type of plywood is chosen, it is absolutely essential that the planking thickness be at least 6mm or ¼ inch. Thicker is fine, but definitely no thinner than 6mm.

Over the last year or so, it has been increasingly difficult to locate good quality BS1088 Gaboon (Okoume) plywood. Most suppliers I've contacted recently have only been stocking a Chinese brand, and although it is stamped BS1088, the examples I've seen have contained a significant quantity of voids in the internal veneers.

The best approach is to do some research to discover reputable suppliers who can provide back-up in the event of flaws in your plywood. I purchase BS1088 ply made from Okoume or Meranti, both of which have interlocking grain which goes a long way towards limiting surface checking. One brand I particularly like is Dragon Hydrotek.

On the plus side, the importers and wholesalers seem to be becoming more choosy about quality due to market pressures. I have noticed an improvement in the general quality of imported "BS1088" ply, but you must inspect all sheets before taking delivery.

I'm not an expert on the legalities of plywood stamping, but I do believe that BS1088 is only enforceable in the United Kingdom, and a Naval Architect friend tells me that BS1088 has been out of force for about ten years anyway.

Let the buyer beware, and use good judgment when choosing your supplier.

Timber

Natural timber components can be made from a fairly wide range of species, with the overriding necessity being light weight and good gluing characteristics. For boats of this size, any decent timber with a density around 32 lbs/cu.ft or 512 kg/cu.m which glues well will do the job. In Australia, suitable species which are readily available are, Silver Quandong, Hoop Pine, Oregon (Douglas Fir), Meranti (if it is good-quality), White Cedar, Western Red Cedar (listed in order of my preference). Other species may be substituted, but do your homework before purchasing. Density as low as 26lbs/cu.ft or 415kg/cu.m may be suitable if used sensibly. In particular, I'm thinking of high-grade Western Red Cedar. Timbers used in construction must be dry, and preferably have a moisture content below 12%.

All of the timber sizes are approximate, and I have made them larger than absolutely necessary. Use your own judgment regarding the use of different sizes (depending on what is available from your supplier), but do not make major changes without contacting the designer or other suitably qualified person.

Adhesives

The only adhesive which I consider suitable at this stage (2024) is a marine grade epoxy such as Bote-Cote, System 3, or West System® Brand Epoxy. There are plenty of others but these are ones with which I have had total success. The reason I recommend epoxy is that it is gap-filling in a truly structural sense, and the ease with which fillets can be formed adds significantly to the strength and durability of the boat. The same resin/hardener combination, used without thickening additives can be used for any fibre-glassing which you find desirable. If you are not familiar with epoxy use, consult a good manual such as "The Gougeon Brothers on Boat Construction" (readily available from retailers and public libraries). Do not compromise when it comes to the selection and use of epoxy - it is what is going to hold your boat together in the long term!

Fastenings

Few permanent fastenings are required in a boat built using the glued-lapstrake (clinker) planking system. Wherever fastenings are used and left in the boat, I strongly recommend silicon bronze ring nails or silicon bronze screws. Do not substitute stainless steel, as it is subject to crevice corrosion. Never use brass, as brass fastenings break easily, and are subject to de-zincification in a marine environment.

Extensive use is made of temporary fastenings. For these, I use stainless steel self-tapping screws most often. These are cheap, easy to drive, and if one should break off within a section of the boat, at least the part left in the timber is protected from corrosion to some extent. To remove screws which have been bonded by cured epoxy, try heating the head for a few minutes with a soldering iron. The heat travels down the screw due to conduction, and the strength of the bonding epoxy is temporarily reduced - the screw will then back out easily. My favourite silicon bronze screws incorporate a square drive instead of a Phillips head or slotted head. You will find that square drives work much more reliably than the other types, and there is no tendency for the driver bit to cam out of the screw head under load.

Gluings

Surfaces which are to be structurally bonded with epoxy must be clean and dust-free. Any shiny surfaces must be well roughened with coarse abrasives to provide an adequate mechanical bond. Oily timbers should be swabbed with a solvent such as acetone to remove oils from the bonding surface. In this case, allow enough time for the acetone to evaporate from the timber, but then prime the surface with un-thickened epoxy/hardener mix to allow the epoxy to penetrate the surface before the natural oils migrate back into the timber fibres.

Measuring

Pay close attention to accuracy when marking and cutting. Use sharp pencils or fine tipped ball point pens for marking timber and plywood. Good quality tape measures and steel rules are an essential investment. One of my favourite measuring tools is a good quality vernier calliper. You don't need a fancy electronic type - just a well-made mechanical tool.

Safety Considerations

Many of the materials used in boat construction are considered toxic. Wood dust is classified as a carcinogen. Equip yourself with a high-quality dual-cartridge respirator and cartridges which are rated for organic vapours (P2 is the rating I look for at Australian suppliers, but make sure to check with your local authorities before deciding on protective equipment). I use my respirators whenever dust and/or solvent vapours are present. Of particular concern is the way in which light weight powdered additives used with epoxy systems get airborne during mixing operations.

Be prepared to purchase plenty of disposable protective hand gloves. I buy half-a-dozen boxes of a hundred at a time, and will frequently change gloves four or five times in a session of work. I am fortunate in that I have not yet become sensitized to epoxy, but many people do after a relatively short time.

I understand that a long list of warnings such as the above can be tedious, but the advice given is based on years of experience – mine and that of many others. Look after your body, so that you can live to enjoy using the boat!

Assembly Instructions for

Phoenix 15

Although both *Phoenix 15* and the original *Phoenix III* have hull shapes and sizes which are very close in size and proportion, they are assembled in a different manner from each other.

Phoenix III is assembled in what could be called the conventional – or even traditional – way, with bulkheads, frames and molds set on a rigid and level bench referred to as a ‘strongback’. All for the cross-sectional components are very carefully set at distances above the strongback, level and at right angles to the centreline, at specific spacings longitudinally and vertical. Making a strongback and establishing the correct location for all the parts is a time-consuming exercise, requiring considerable skill and attention to detail.

Once the strongback and the framework for the boat are satisfactorily in place, the shape of individual planks must be determined using a procedure called ‘spiling’. Many amateurs have mastered this technique, but once again, it requires substantial material and lots of time.

When we began discussing the possibility of producing CNC-cut kits for *Phoenix 15*, reducing the cost of materials, minimizing assembly time, and simplifying the strongback were all important objectives. We believe that what we have come up with has exceeded our expectations.

Not only have all of the plywood components been accurately cut on the factory floor by Computer Numerically Controlled routing machines, but the transverse and longitudinal parts fit together using small mortis and tenon joints – i.e. tabs and slots machined into the plywood parts. This approach removes most of the measurement and centering normally required when setting up the boat, but it virtually guarantees an accurate and symmetrical boat.

For those who decide to build from scratch using plans, the benefits of the simplified construction methods still apply. There is no need for spiling to establish the plank shapes because I have included offset dimensions which allow plank shapes to be marked directly on the sheets of plywood. Similarly, there are dimensions supplied for frames, bulkheads, centerboard and rudder etc.

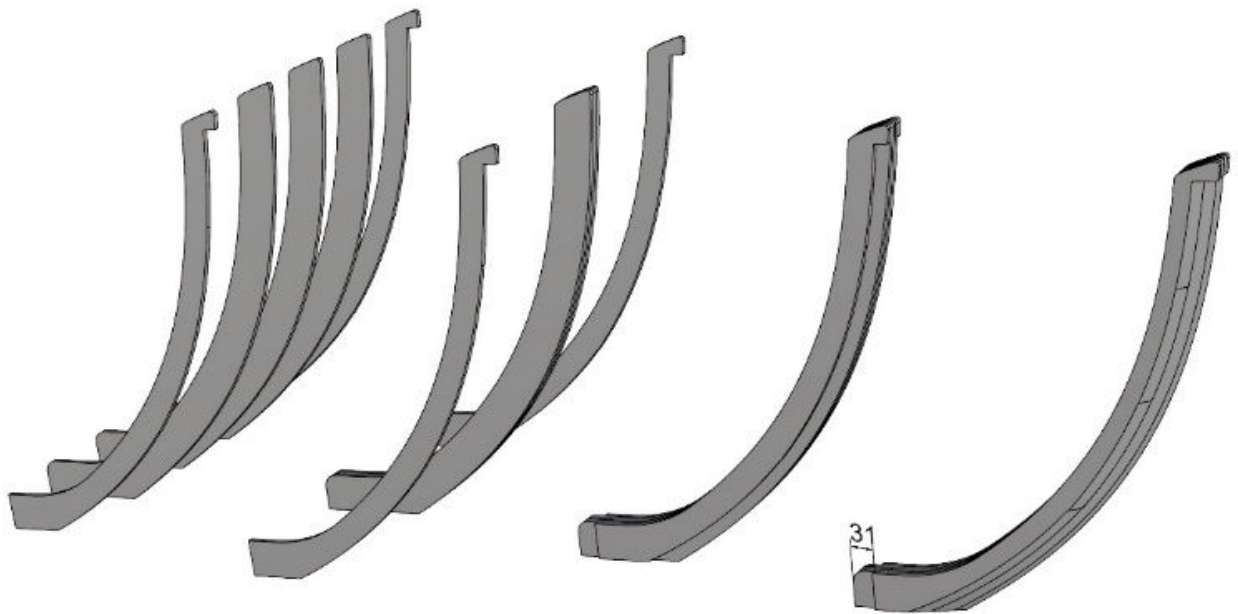


Basic Building Sequence

Sub-assemblies

Stem

It is a good idea to start with the gluing-up of sub-assemblies which will be required prior to the commencement of construction. Let's begin with the inner stem of the boat.



There are five pieces of plywood required for the inner stem, all five glued together using thickened epoxy. There are three pieces of identical shape which make up the central core of the stem, two of which should have a fine score line on one of their faces. These score lines are used in determining the location of the remaining two stem laminations, so dry assemble the three core laminations with the outer two being the pair with the score line, and the score lines facing outwards.

The remaining two laminations also should have a score line on one face. Dry assemble the pieces with the score lines facing outwards. These score lines will help you enormously when cutting the final bevel to accept the planking on the glued-up stem assembly.

When you are satisfied that you understand what the finished, glued assembly should look like, open out the laminations on a flat surface and apply epoxy to the mating surfaces. Remember that the two outer laminations will only need glue on one face each, which will be the face without a score line.

When using epoxy as an adhesive, it is best practice to apply a priming coat of unthickened epoxy to the joining (faying) surfaces. This allows the low-viscosity resin/hardener to penetrate the wood surface and provide an excellent bonding surface when the thickened adhesive batch is applied.

Wet epoxy is slippery, so unless you are an experienced builder, I recommend that the stem lamination be done in two stages – first the three inner laminations, and after they have cured and been cleaned up, prime, glue and apply the two outer laminations.

Use a try-square or a series of right-angled blocks attached to your building surface to ensure that the laminations always remain correctly lined-up. The lubricating qualities of the wet epoxy make it quite difficult to do by eye and feel. For the uninitiated, I recommend that when the pieces are dry assembled, pilot holes be drilled at several locations and temporary screws be fitted. Make sure the screws penetrate all layers to prevent any slipping out of alignment. Once again – do this dry.

Whether you use the temporary screw method or some other set-up using blocks is up to you, but keeping the laminations true is important for a good result.

Epoxy does not require high clamping pressure – contact pressure is enough. The temporary screws can do double-duty as the clamping system if they are evenly spread out along the length of the stem – first the inner three-piece lamination, followed later by the two outer laminations.

The use of temporary screws in wood-epoxy building is helpful in all sorts of situations. Be careful to remove the screws before the epoxy has cured too much (the time depends a lot on the ambient temperature in your building space). Leave them in long enough for the lamination to hold, but not so long that they become glued in and break when removal is attempted.

I try to use 316 stainless steel CSK self-tappers so that if a screw does break during removal, at least the metal left in the joint is reasonably corrosion resistant.

If the screws are too difficult to remove safely, I apply the squared-off tip of a soldering iron to the head of the screw for about a minute to allow the heat to be conducted through the metal. Then it is usually easy to back the screw out safely.

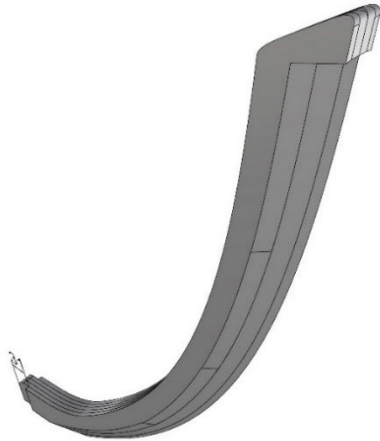




After the laminated stem assembly has cured, remove any remaining screws or clamps, fill holes with epoxy mixed with lower density sanding filler additive or wooden dowels/toothpicks etc., and scrape epoxy squeeze-out and drips. The lower ends of the two outer laminations should finish 31mm short of the lower ends of the three inner laminations as shown below.



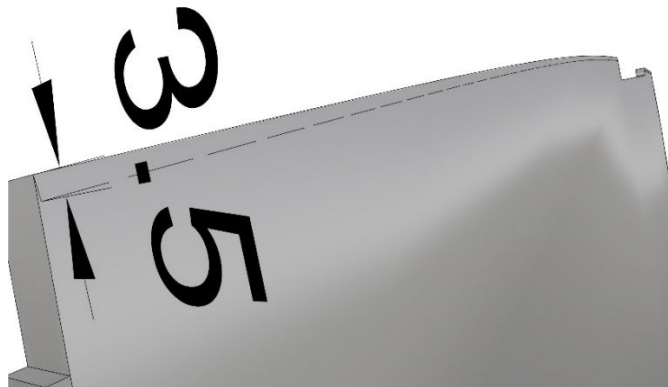
The final stage of stem construction is to make a bevel which runs between the score line on the outer laminations and the centre of the face of the stem. It will make it easier if you put a bold centreline on the forward face of the stem using a soft pencil, a ballpoint pen or a felt tip marker. Cut the bevel on either side of the stem from the score line to the centreline on the face. This is best done with a spokeshave, plane, and/or rasp and chisel. Some people may prefer to use a belt sander, but I find them to be unpredictable, dusty, and tricky to use.



If you can follow the score line, and bevel down until just short of the marked centreline, the bevel will be correct. Take it down almost to a sharp feather edge but no further. Just leave a little bit of the flat face remaining – say about 2mm.

Transom

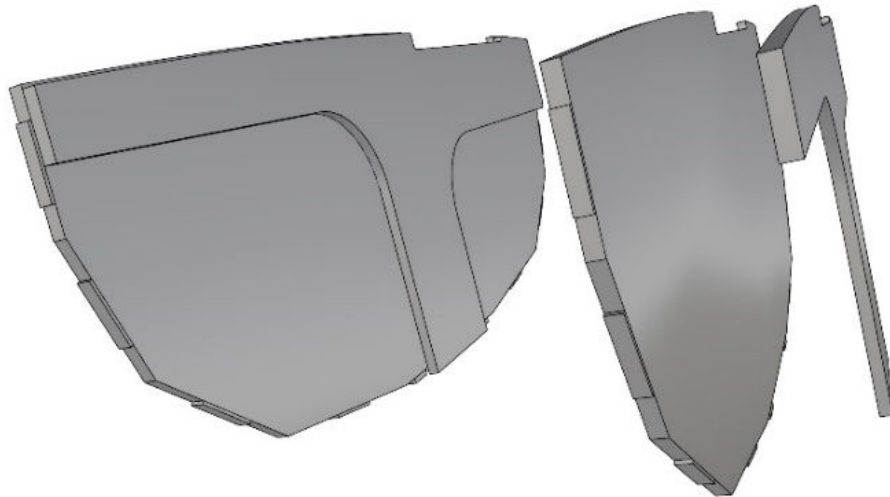
The transom is made from two pieces of 12mm plywood – the transom itself and an internal doubler. Mark an line on the inside face of the transom plywood, 3.5mm parallel with the upper edge as shown below.



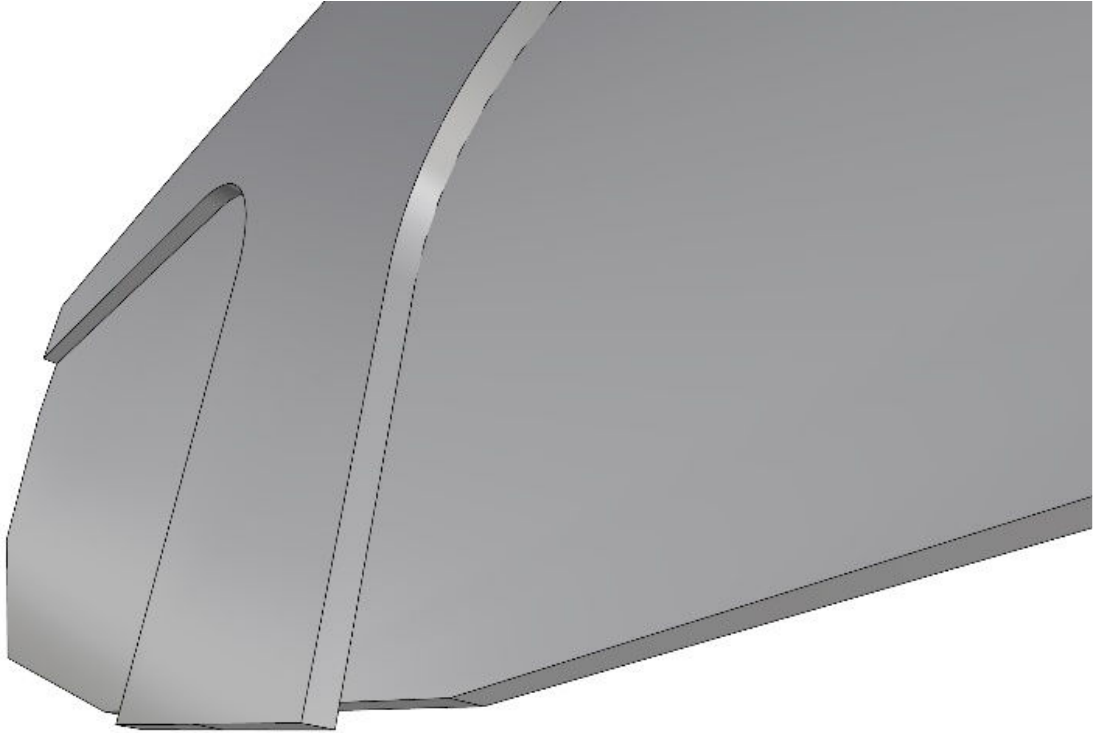
Position the doubler with its top edge against the line just drawn. Mark around the edge of the doubler onto the transom face to show where glue is required during assembly. Prime the face of the doubler and inside the marked faying (i.e. joining) surface as marked on the transom.

Apply epoxy thickened with adhesive powder additive and assemble using liberal numbers of temporary screws or nails. Be careful to apply enough thickened epoxy mix so that there are no air gaps. Clamps can be used to apply pressure, but they tend to apply pressure around the edges only. In any case a few locating screws are recommended to prevent the two laminations from sliding out of alignment.

Although it is not practical for a one-off job, vacuum clamping (i.e. vacuum bagging) is the best way of applying even clamping pressure over large surface areas. In any case, temporary locating screws should be used. I use polymer nails fired in with a special pneumatic nail gun – if you have the facilities, this is a wonderful way of locating and applying pressure, and the polymer nails can remain in the job forever. You can sand, drill or cut right through them without damaging tools.

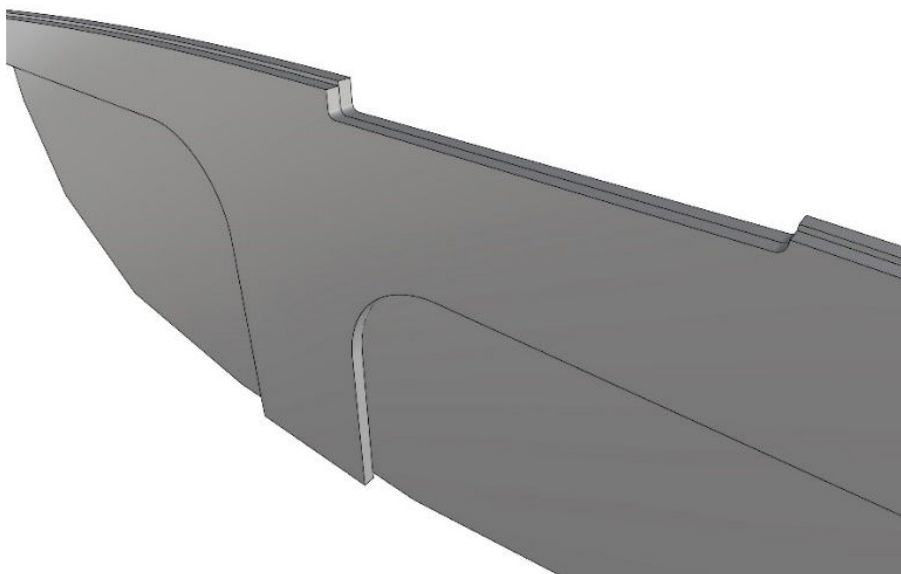


When the lamination has cured the transom and doubler should look like this...

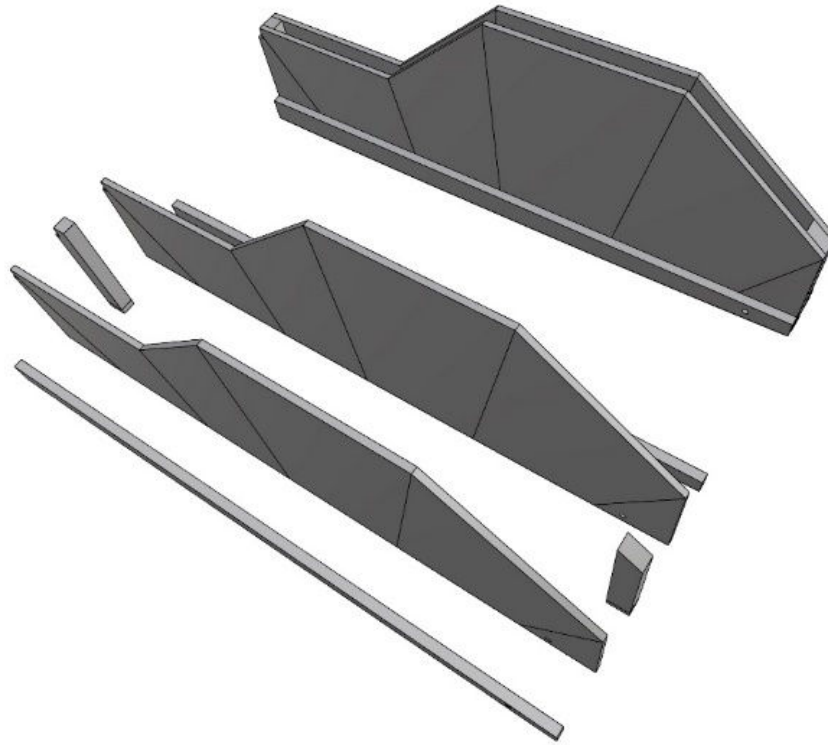


...with the lower end of the doubler protruding beyond the edge of the transom by about 8mm (5/16"). The protruding part of the doubler needs to be bevelled at an angle of 19 degrees as shown in the next image

The upper edges of the assembly will appear staggered as below.



Centreboard Case

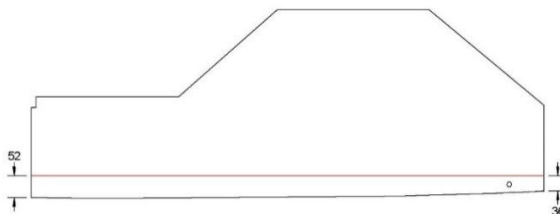


The above image shows the six main parts of the assembly, with the assembled case in the background. The six parts are:-

- Two plywood sides;
- Head ledges (i.e. end posts);
- and two longitudinal bedlogs.

I recommend covering the inside faces of the plywood with 200gsm (6oz) woven glass cloth set in epoxy to protect against wear and moisture intrusion. Once the case is assembled it will be difficult to work on the inside surfaces.

First thing to do is to mark the outside face of each plywood side for positioning of the bedlogs. The bedlogs are made from 19mm ($3/4''$) thick timber at least 56mm ($2-3/16''$) wide. I use 19mm x 65mm (about $3/4'' \times 2-1/2''$) as it is a standard dressed size where I live. Mark a line on the outside plywood faces as shown below:-



Clamp your selected timber with the top edge against the marked line, and then trace the curved bottom edge of the plywood onto the

bedlog. Do the same on the second one and then trim them to shape using a jigsaw, bandsaw or a fine-toothed saw such as a Japanese pull-saw.

With bottom edges of the bedlogs flush with the bottom of the case sides, drill and drive a temporary screw through both ends of both case sides. This is so that the bedlogs will remain in the correct position when you glue them and the plywood sides together.



The photo above shows the case sides being glued to the shaped bedlogs. The case sides are blocked up with off-cuts so they will remain parallel with the surface of the bench. Even though I've used a lot of clamps, the clamping pressure is low, with the aim being to just provide even contact pressure. You can use temporary screws or even lead weights to apply pressure, but make sure that the bench is true so that everything comes out straight.

If you intend sheathing the inside surfaces of the case sides with glass or Dynel, this is the time to do it.



Cutting glass cloth to fit sides. Leave about 12mm (1/2") overlap all around.



I use a soft paint brush or dustpan brush laid over at about 45 degrees to smooth out the loose weave of the cloth.



Pour on unthickened epoxy and spread evenly. I prefer to use a squeegee – in this case a piece of 1.5mm (1/16") model aircraft Balsa. Work gently and slowly and be careful not to aerate the epoxy. The aim is to get enough epoxy on to just wet-out the cloth and no more.



Spend time to gently even out the surface, removing any excess epoxy. I use squeegees and dry disposable paint brushes.



This is the surface – fully wet-out, but with no excess resin. I recommend waiting until the first layer of epoxy has cured enough to ensure the glass is partially adhered to the plywood, and then apply a second coat to fill the weave. This should be done within the 'chemical bonding window' which in hot weather should be not much more than four hours, and up to twelve hours in cooler weather. After the second coat has become tacky, a third coat should fully fill the weave of the cloth. After full cure, sand the surface smooth, but don't sand so deep as to cut the glass.



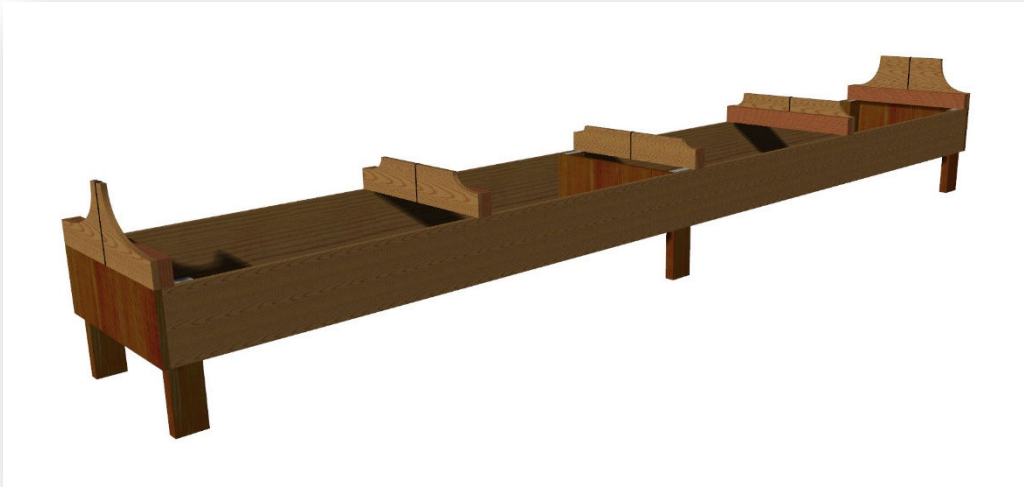
Once the glass/epoxy has cured, glue and screw the head ledges into position, making sure the edges facing the inside of the case are well coated with epoxy to protect against water soakage. In the photo above you can see that even though the head ledges and case sides are joined using epoxy and silicon bronze screws, I have clamped dry head ledge offcuts lightly inside the slot to help keep the inner faces of the case opening parallel.



Note the nice squeeze-out from the joint and the heads of the silicon bronze screws.

Strongback

The strongback should be made to the dimensions supplied with the kit or plans, although you



may wish to substitute materials which you have on hand. No matter what it is built from, it is important that the strongback is strong and rigid, and preferably bolted or otherwise firmly attached to the floor. The cross supports are attached to the strongback at longitudinal distances as specified and made to exactly the heights shown in the documentation.

When all is done, take plenty of time to sight along the tops of the cross supports to make sure that there is no twist in the structure, and double-check all the measurements. It is particularly important to ensure that there is a centreline on the faces of the five cross supports and that they are all in a straight line. Your eyes will do a good job of sighting along at a low angle, and you will be able to spot any which are out of line. A stringline or a long straightedge will do the job also.

Centreboard Case and Stem Sub-assemblies

Draw a straight line on the under surface of the bottom panel running from the tip at the forward end to the centre of the aft end. A good way to do this is to lay the panel on a flat floor and 'snap' a chalk line. Alternatively, use an accurate straight edge or set up a string line and mark carefully with a pencil. Included with your kit is a detailed drawing of the locations of the cross supports, their heights, and the amount that the bottom panel overhangs the first cross piece at the forward end.

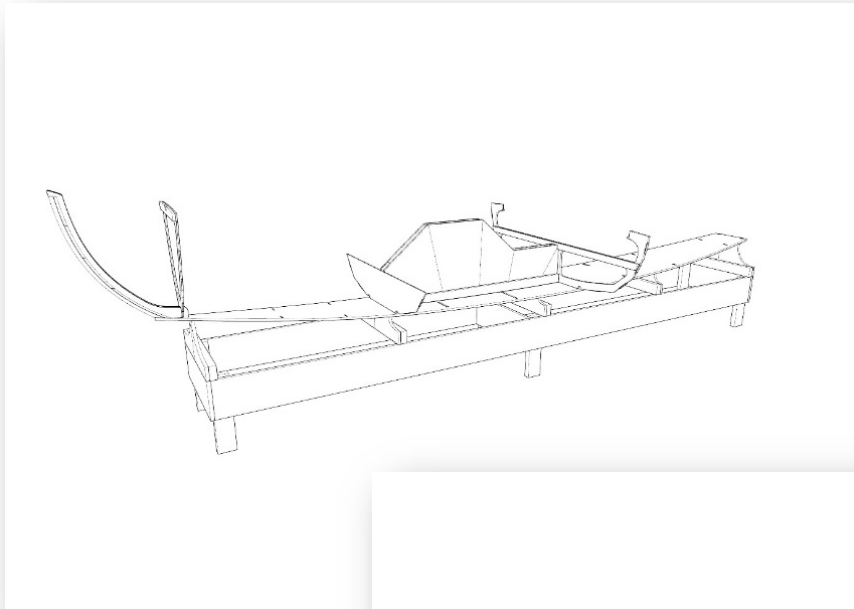
Put some temporary weights on the bottom panel so that it sits in contact with the cross supports and check underneath the panel to ensure the centreline lines up with the centrelines on the cross supports. When all is correct, use temporary fasteners to tack the bottom down firmly.

Assemble the centreboard case according to the instructions and place into position on the bottom panel without any glue at this stage. The protruding head-ledges (end posts) should slip into the slot already cut out of the bottom panel. Dry fit (i.e. without any glue) the centreboard bracing bulkhead at the forward end of the case, and the midships frame – station number 6 –

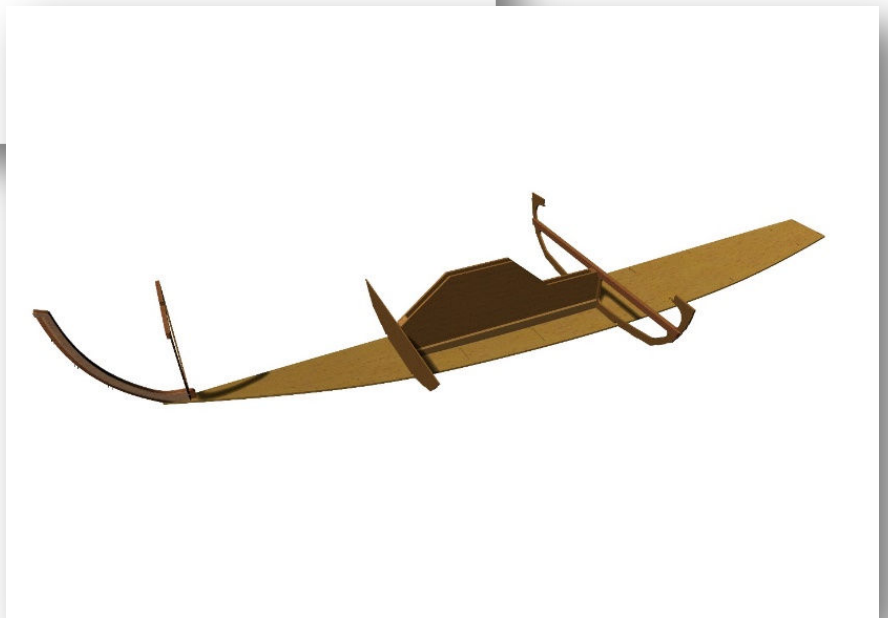
at the aft end. The tabs, or 'tenons' on the frame bases will fit into the mortises which have been machined into the bottom panel. Trim or bevel the tenons a little if required to fit the mortises. The bracing bulkhead and the midships frame will be held vertical by the centreboard case, but use a try square to ensure the case itself stands vertical. When all parts are correctly aligned, drill for and drive screws through the bulkhead and frame into the end posts of the case (more details in the detailed instructions).

Drive screws up through the bottom panel along either side of the centreboard slot up into the bed-logs (the larger timber longitudinals on either side of the centreboard case) at the bottom edge.

With the centreboard case, bracing bulkhead and the midships frame all screwed together and the bottom panel screwed to the centreboard case assembly it is time to double-check that all components are sitting vertical in the appropriate planes. When any adjustments have been



made and you are happy, remove the screws and commence permanent assembly. Prime all the faying surfaces ('faying' means the surfaces which meet each other – in this case the surfaces which will be glued together) with un-thickened epoxy resin/hardener mix, and then apply thickened epoxy/hardener mix to at least



one surface of each pair of primed, faying surfaces and screw the complete assembly into its final location.

Use a tongue depressor or similar to create neat epoxy fillets with the squeeze-out and remove the excess before it cures. Check and re-check that the components are vertical and that the bottoms of the bracing bulkhead and the midships frame are in even contact with the bottom panel. If not, use clamps, weights and/or temporary fastenings to achieve even contact.

The laminated stem and frame #1 can now be fitted using the same sequence of operations (dry fitting with screws through the bottom panel into the base of the stem assembly, and temporary fastenings to hold frame 1 to the stem). The outer cheeks of the stem will help in locating frame #1 correctly. Improvise as necessary to ensure that the components are vertical and that the forward point of the bottom panel doesn't droop due to the weight of the stem. Improvising and planning are very important when boatbuilding and add greatly to the creativity and satisfaction derived from your project. Do not be afraid to ask for clarification from Duck Flat Wooden Boats, and make sure you read as many books on boatbuilding as you can manage. Glue as previously described for the centreboard-case, brace bulkhead and midships frame.

Your boat should now be looking like the drawings on the previous page.

First Planks

Select the pair of planks which adjoin the bottom panel (these are plank #1, port and starboard) and lay them on a flat surface (floor, bench or sawhorses) with the mortises (the cut slots) facing upward. Note that the planks are mirror images.

Drill a series of holes along the lower edges (the edge which will touch the outer edge of the 12mm bottom panel) of the planks to take cable ties or copper wire stitching. These holes should be 8mm (5/16") in from the edge and spaced at approximately 100mm/4in centres. Depending on whether you use plastic cable-ties or copper wire, the holes could be anywhere from 2mm to 5mm in diameter. The resulting hole does not damage the hull because during the building process, the holes will be filled with epoxy and covered with glass cloth set in epoxy. Remember, these holes are only required along the lower edge of plank #1. The lower edge is



the shorter one due to the curvature of the forward end where it meets the stem. Avoid drilling holes close to where the frames and bulkheads will lie – you can see this from the mortises (slots) in both the planks and the bottom panel.

Get an assistant to hold the forward end of the plank up near the stem while you hold the plank close to the midships frame which had be glued on to the bottom earlier. Locate the tenon (protruding tab) on the midships frame in the matching mortise (slot) in the plank and place a spring-clamp or a bar-clamp to temporarily hold the plank up against the midships frame. Move



forward and do the same thing at the brace bulkhead and frame #1. You will not be able to clamp at the brace bulkhead because it is a flat surface. I would hold the plank against the edge of the brace bulkhead with the fingers of one hand, ensuring that the lower edge of the plank is snug against the outer edge of the bottom panel as well. Lightly tack or screw through the plank and into the edge of the brace bulkhead, being careful because the bulkhead is only 6mm thick. Another way of doing it – one which I prefer – is to drill two holes about 8mm (5/16”) below the upper edge of the plank on either side of the brace bulkhead and a hole through the bulkhead adjacent to the plank holes as you can see in the photos below (these photos are from another boat – in your situation the brace bulkhead will continue above the plank edge, but the method is the same). Run a cable-tie through as per the photo and lightly draw the plank edge in against the brace bulkhead.

Tack, screw or clamp the forward end of the plank against the bevelled face of the stem. Exact positioning is not critical now.

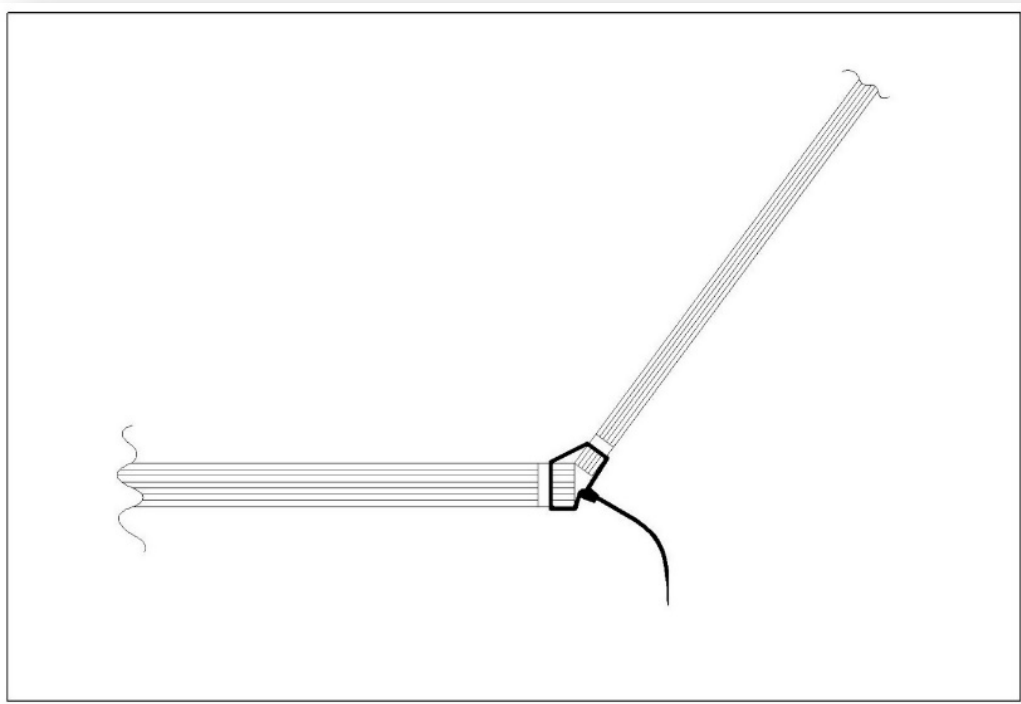
Starting in the area between the brace bulkhead and the midships bulkhead where plank #1 is clamped or cable-tied into position, drill a series of holes 8mm (5/16”) in from the edge of the bottom panel with each directly opposite the existing holes along the bottom edge of plank #1.

Using cable ties or copper wire, stitch the joint full length.

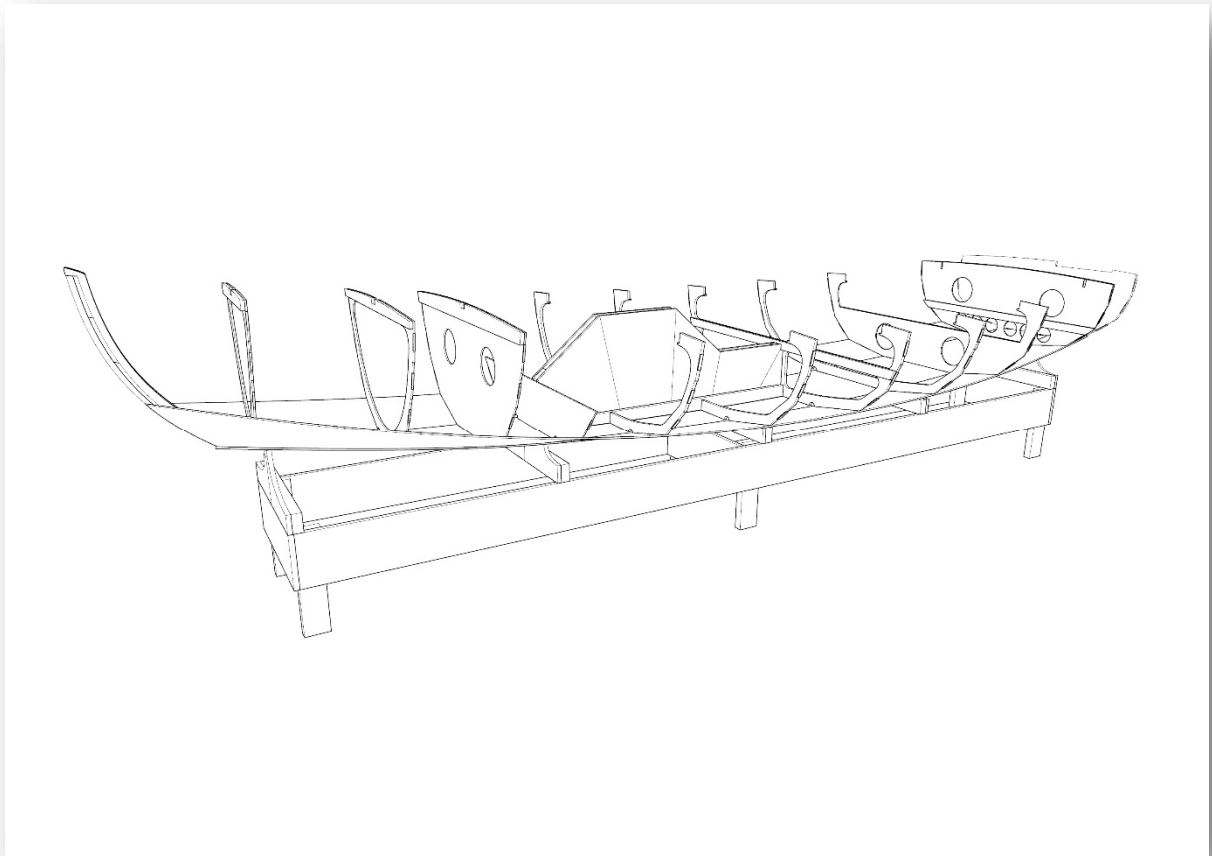
If you look carefully at the photo below you should be able to see cable-ties along the adjoining edges of the bottom panel and the first plank. (This is a different design – the boat in this photo has the frames and bulkheads go in after plank #1 is installed).



Run cable-ties through the holes in the bottom edge of plank #1 and those in the outer edge of the bottom panel. The sketch below is a cross-section through the bottom panel and plank #1, showing the alignment of the panels and the way in which the cable-tie is positioned. The alignment of the plywood panels is helped if the sharp corners which come into contact are sanded or lightly planed to take off the sharp corner. Just a little is required.

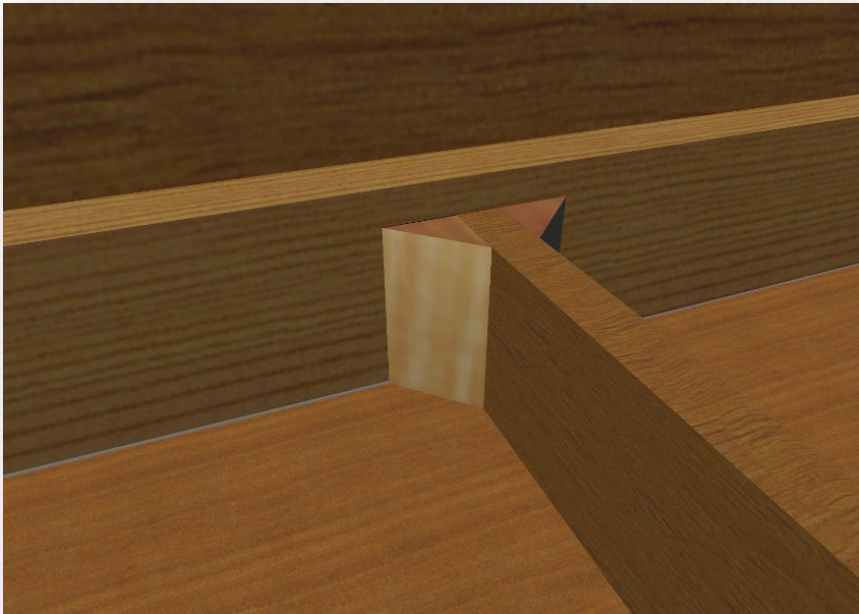


Fitting the remaining frames and bulkheads



Starting with frame #2 and moving aft, dry fit all the remaining frames and bulkheads, using clamps, cable-ties or temporary screws and nails. Take your time over this stage of building, ensuring that the components (not the transom – it comes later) are vertical and that there is no twist in the boat.

Frames #4 and #5 are in port and starboard halves because of the location of the centreboard case. The inboard ends butt up against the centreboard case bed-logs (i.e. the timber longitudinals at the base of the CB case) and when the time comes to glue them into their permanent location the joints between the inner ends and the bed logs should be reinforced by small timber blocks of 19mm x 19mm or larger cross-sectional dimension. Large epoxy fillets are another alternative, but if used, I recommend the coved surface be covered with a strip of woven glass extending out onto the frame face and the bed log. See image on following page...



Using the 'Stitch-and-Glue' method to attach Plank #1 to the bottom and Frames/Bulkheads

Now that the frames, bulkheads and plank #1 are all joined together with plastic cable ties (my preferred method) or by twisted copper wire, work over the structure gently confirming that everything is in alignment and curves are fair and smooth. There is no hurry, and you can tighten, loosen, or even remove and replace individual ties until the boat looks fine.

When the fellows at Duck Flat Wooden Boats were helping students build a number of Phoenix 15's during a two-week course, they tried an interesting method to help with frame and bulkhead alignment. Taking advantage of our 'Mortise-and-Tenon'-style kit, they temporarily attached the fourth plank (i.e. the Sheer Plank) to the heads of the frames and bulkheads. The tenons on the frames and bulkheads mated up with the mortises in the fourth plank to keep the frame heads true.



In the above photo the boys are applying epoxy fillets to the joint between the bottom panel and the lower edge of plank #1. Note how plank #4 has been dry-fitted using the mortise-and-tenon joints. People building from scratch using plans won't have mortises and tenons, but the

same process can be folled by reference to the frame and bulhead markings made using dimensions shown on the drawings.

Now, back to the business of fitting plank #1 to the outer edges of the bottom panel.



I use small 'Zip-Loc' bags with one corner cut off to act like a cake icing bag, when laying down small epoxy fillets.



Here you can see thickened epoxy being laid down. This is what will end up being a fully structural fillet – note that I have primed the adjoining surfaces with unthickened epoxy to ensure the adhesion of the fillet when smoothed out.



When joining plank #1 to the bottom, I recommend doing a series of ‘tack welds’ remaining clear of the plastic ties. These fillets are small enough to be completely covered by the heavier structural fillet which comes later, but to make sure I generally press and flatten them down a bit using a tongue depressor.



Note in the image below that the 'tacks' have been flattened out somewhat and are small enough to be covered completely when the final fillets and the glass tape goes on.



While the 'tack-weld' fillets are curing, I take the time to pre-cut the glass tapes (this boat is *Flint*, not *Phoenix 15*.)





After the 'tack-welds' have cured for about 12 hours, I carefully remove the cable ties and pre-wet the surfaces with un-thickened epoxy, for the taping.



In the above photo, you can see a fillet of thickened epoxy roughly laid into position. It should be fairly even, but you don't need to be obsessive, as the glass tape will smooth it out.



Next, lay the pre-cut tapes into the wet fillet.

Press lightly to hold it in position but avoid depressions from your fingers.



Brush un-thickened epoxy through the glass until fully wet-out - but use no more than necessary. A commercial compression roller is handy for getting an even, bubble-free finish.



Here is *First Mate* with all the internal glass taping completed. In the case of *Phoenix 15* the only joints you will be doing at the moment are between the bottom edge of plank #1 and the flat bottom panel, plus on either side of all of the frames and bulkheads where they lay against the bottom and plank #1.

Having spoken about glass taping the bulkheads and frames into position, you can save much labour by using the glass tape only on the hull seams and relying on just fillets of thickened epoxy to secure the transvers components. Refer to pages 101 – 105 in the following superb book <https://www.westsystem.com/app/uploads/2022/10/GougeonBook-061205-1.pdf>

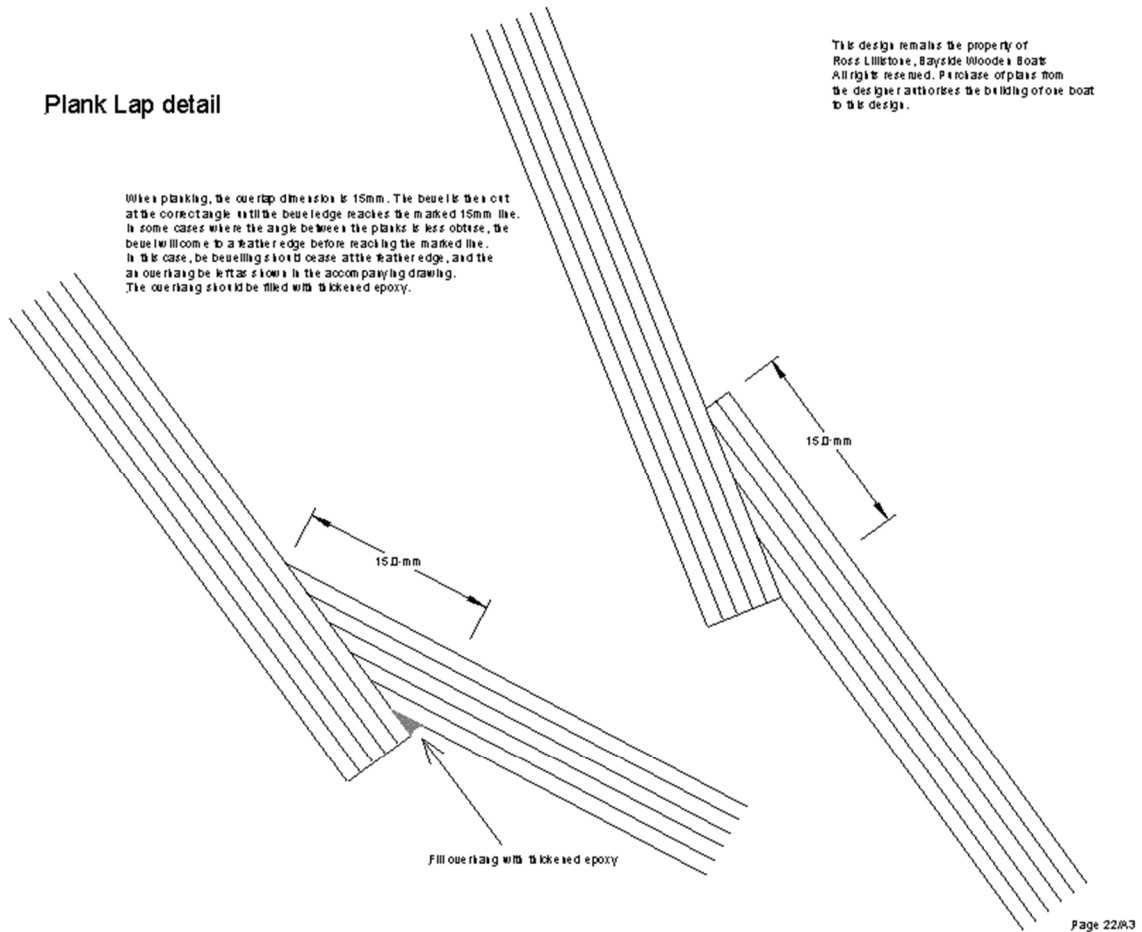
Bevelling the Plank Laps

Now that the first (garboard) plank is in position, it is time to cut the bevel so that the second plank will lie in full contact. See drawing below. (Note: this drawing is viewed as if the boat is right side up.)

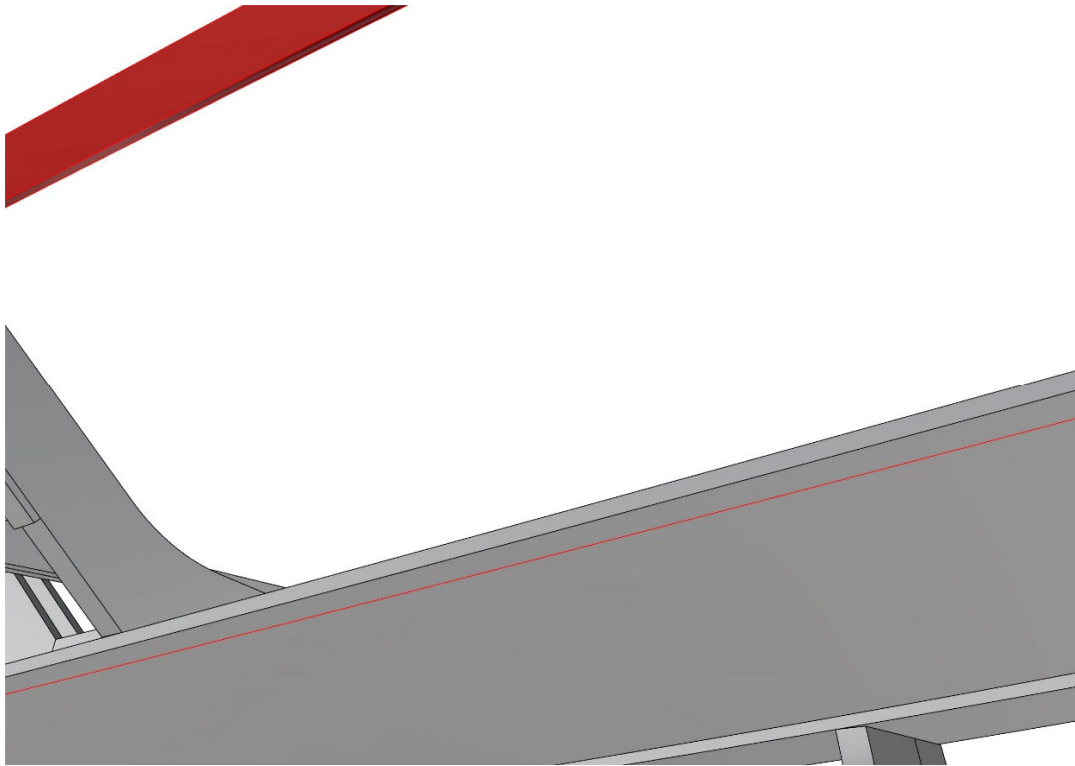
Plank Lap detail

When planking, the overlap dimension is 15mm. The bevel is cut out at the correct angle until the bevel edge reaches the marked 15mm line. In some cases where the angle between the planks is less obtuse, the bevel will come to a knatter edge before reaching the marked line. In this case, beveling should cease at the knatter edge, and the overlap dimension shown in the accompanying drawing. The overlap should be filled with thickened epoxy.

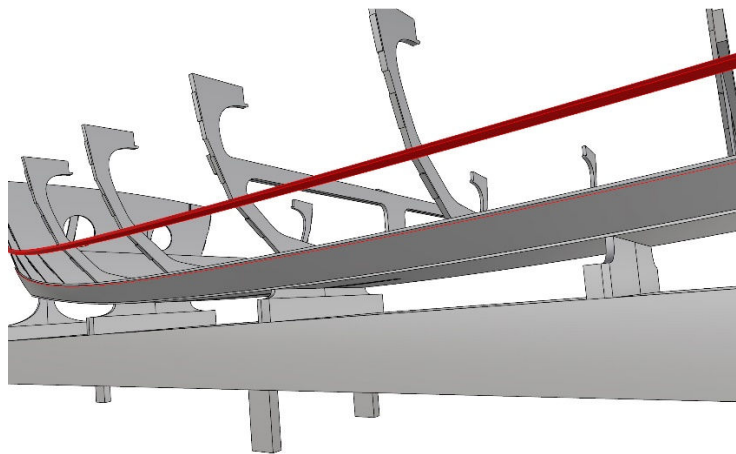
This design remains the property of
Ross Littleton, Bayside Wooden Boat
All rights reserved. Permission of plans from
the designer authorizes the building of one boat
to this design.



In the case of *Phoenix 15*, the width of the overlap is 15mm/5/8". Mark a pencil or ballpoint line parallel to, and 15mm (5/8") down from the upper edge of plank #1. This line defines the limit of the overlap between the planks, and therefore, defines the area which will be bevelled.



Above you can just see the red line marked 15mm (5/8") in from the outer edge of the garboard strake i.e. plank #1.



Next, attach a batten to the molds and bulkheads with its outer edge aligned against the next plank mark. This will allow you to cut the bevel with a constantly changing angle

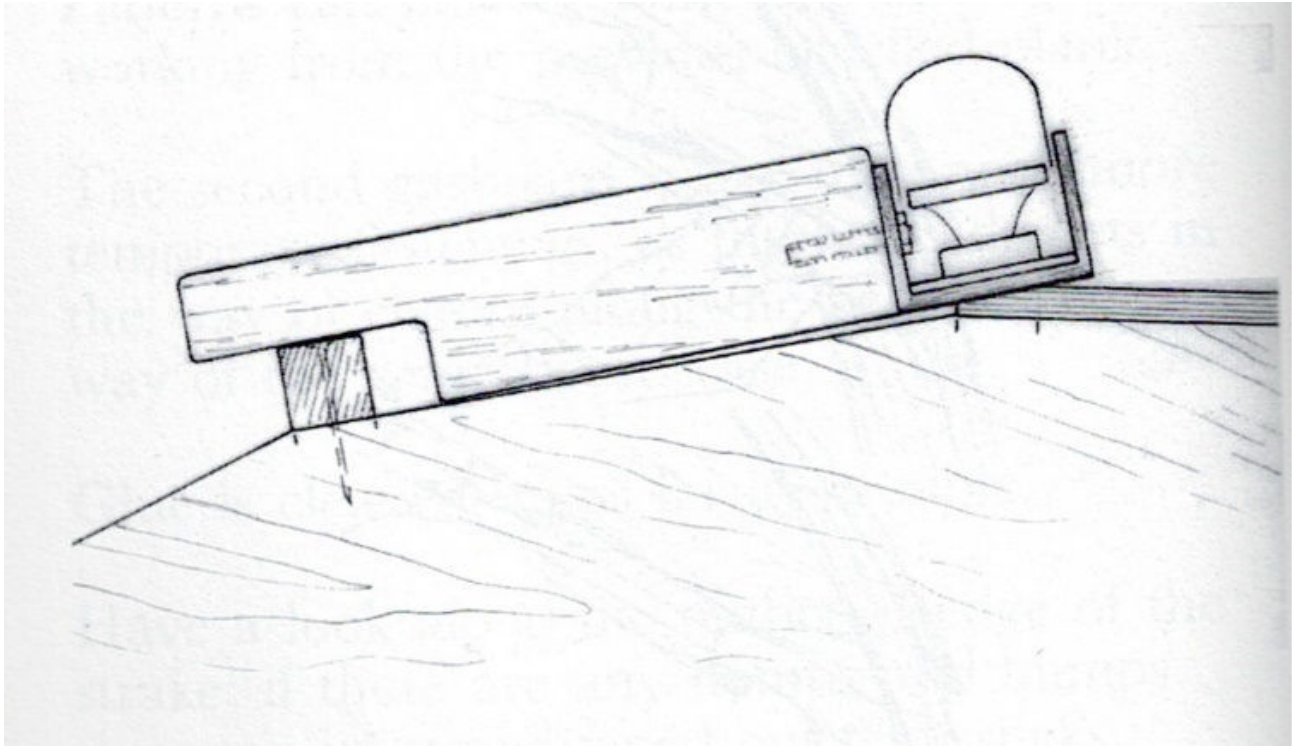
automatically and get an accurate result every time. I make use of a modified block plane and cut the bevel until it reaches the marked 15mm/5/8" line.



This photo shows me planning a bevel on plank #1 on my *Periwinkle* design. This boat is built upside-down, but the principles are identical. The temporary batten just above my left arm is 12mm (1/2") thick, and the brass rod I've got running through two drilled holes in my block plane is set so that there is 12mm (1/2") between the sole of the plane and the lower edge of the rod. There are many ways to set-up your plane for this work.

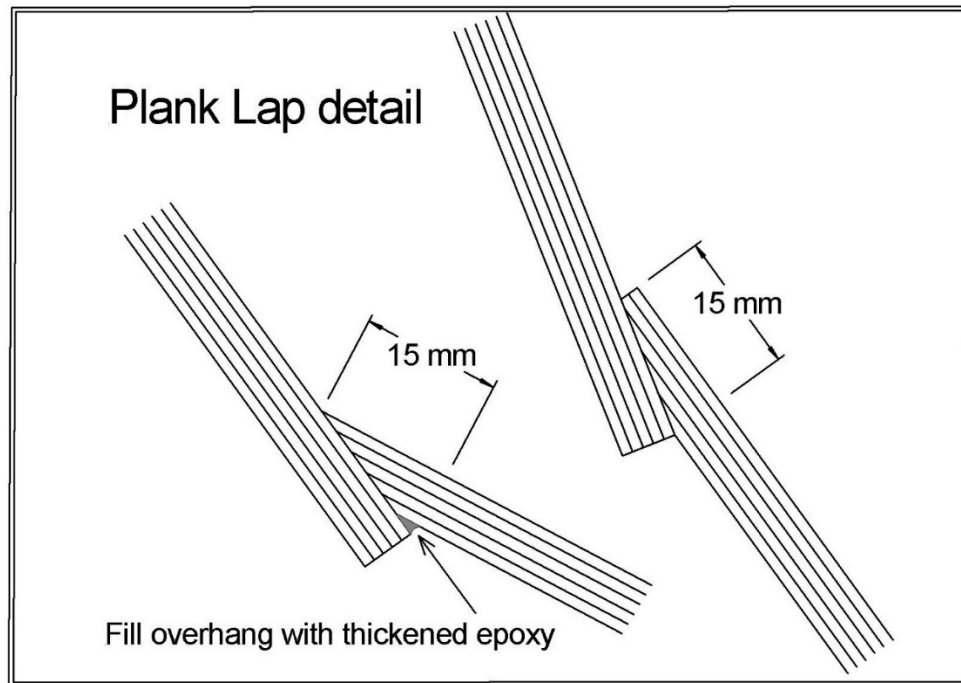


Here is a close-up of the operation – this time bevelling a keelson, but the principle is clear. These days I attach a block of wood which is long enough to span the plank width, using a screw driven from the inside of the plane out into the end of the wood block. That leaves space for the wood shavings to clear the plane where they sometimes used to be blocked by the brass rod.



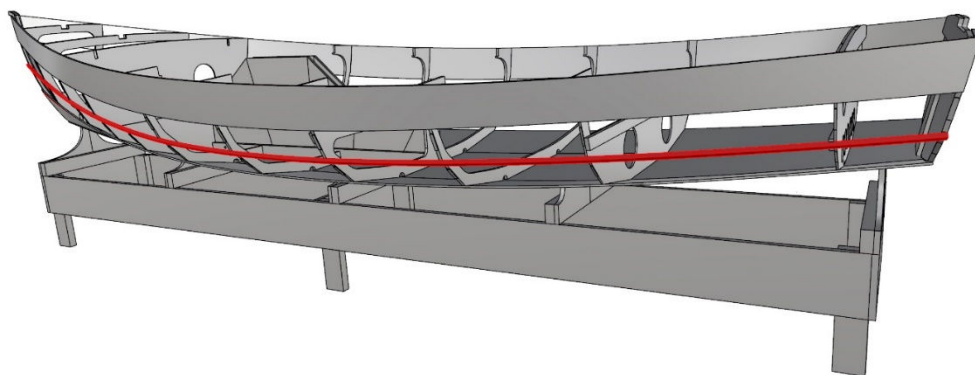
Above is a sketch taken from the late Iain Oughtred's superb book, 'Clinker Plywood Boatbuilding Manual', which I believe to be the best book yet written about glued-lapstrake construction methods. This shows the operating principle for the modified plane.

Most of the time you will just have to plane until the bevel touches the marked line as in the sketch on the right side of the picture below. However, where there is a steeper angle between the planks you may find that the bevel starts to make a feather edge before the bevel gets to the marked line. In that case cease bevelling as soon as you start to form a feather edge. The joining plank will still line up with the 15mm (5/8") marked line, but there will be an overhang which will be filled with thickened epoxy when the plank is hung. See the sketch on the left-hand side of the picture below.



For this bevelling stage the transom assembly should be lifted into position and attached using temporary screws. To aid with the angular positioning of the relatively heavy transom and doubler assembly, I suggest attaching plank #4, once again using temporary fasteners – screws or cable ties run through small holes as was done when hanging plank #1.

The mortise-and-tenon assembly system will aid in keeping everything in-line but keep checking with your eyes and spirit level. See image below.



Once you are happy that the whole assembly free of twist from stem to stern, and that the plank curves are all fair, brush unthickened epoxy into the open V-shaped gaps between the inner side of the transom, the bottom, and the two planks #1. Try to work the resin under the angled lower edge of the doubler, although there won't be much gap there. Capillary action will pull the liquid resin in to a great extent. When the joining edges and faces of the bottom, transom assembly and planks #1 are properly primed, apply a generous thickened epoxy fillet. Remember to leave plank #4 dry and just temporarily attached, as it is only helping to align components at this stage.

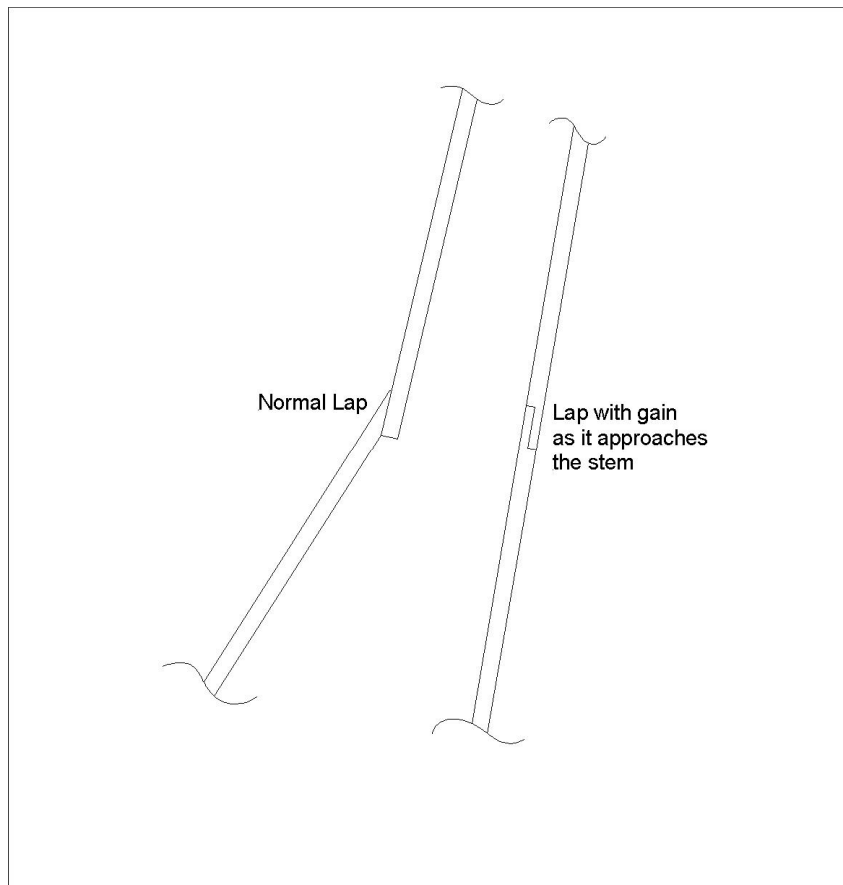
Stand back and check the assembly by eye and spirit level several times more, and if you are happy that there is no twist and that curves are fair, walk away and leave the epoxy to cure overnight.

Plank Gains

Before attaching the second plank to the boat, there remains one more bit of woodworking, and that is cutting the gains.

When lapstrake planking is fixed onto the boat, there is an obvious ridge of overlap on the outside of the boat which is usually equal in thickness to the planking material.

As the planking approaches the stem it is usual to make the overlapping thickness fade out until it disappears





See here how the plank laps disappear totally just before reaching the line between the inner and outer stem.

The way to accomplish this reduction in thickness of overlap is to cut a rebate in the planks over the last 350mm (14") approximately of the overlap area. This rebate is best cut using a rebate plane, or a router. If you don't have these tools, you can do the work with a sharp chisel without any problem

It is possible to cut the full depth of the rebate in a single plank, but better practice is to cut half the depth in one plank and the remaining half-depth in the mating surface of the inside face of the next plank





Plank 2

By this stage you should be ready to attach plank #2 permanently. Plank #1 has been bevelled, and the gains approaching the stem have been cut. Remove the batten used for the plank overlap bevelling and lift plank #2 into position on one side of the boat. Trim the tenons on the frames and bulkheads until they slip into the shallow mortises, and starting in the middle, attach the plank using temporary fastenings such as thin nails or light clamps. My preferred method is to drill through the edges of the plank and into the edge grain of the bulkheads and frames. I then use 6ga or 8ga disposable self-tapping screws to lightly hold the plank in place. It is advisable to use permanent screws at both ends of the planks where they lay on the stem and the transom – I use 8ga silicon-bronze screws of about 31mm (1-1/4").

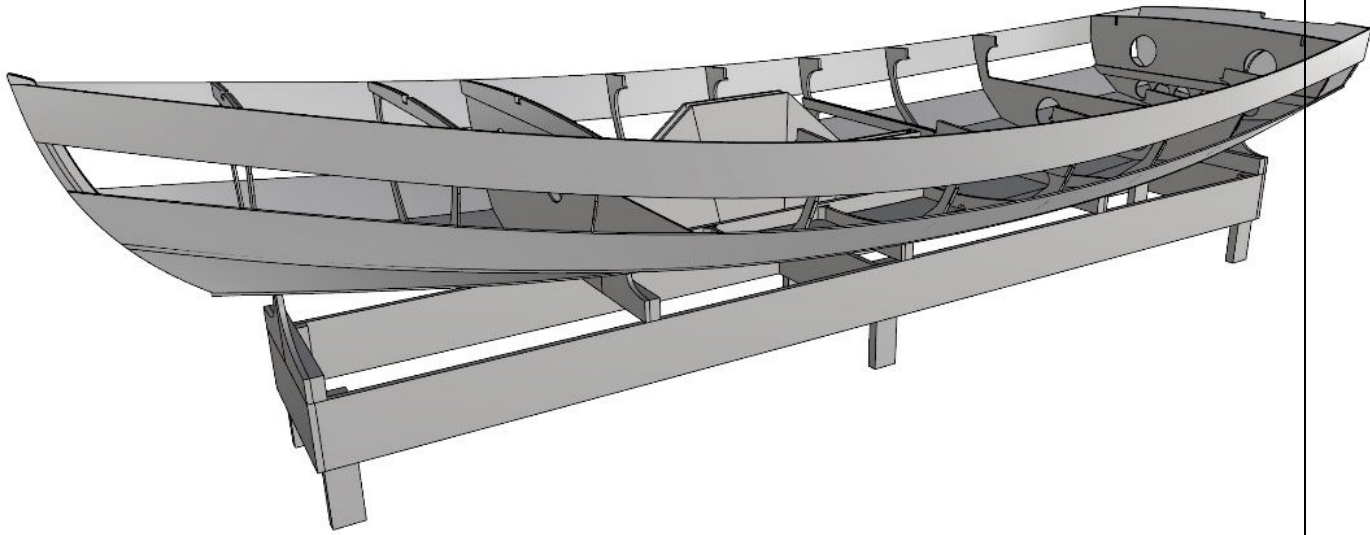
Use your eyes to look along the plank edges from a number of angles to ensure that everything is smooth and fair. Shave or pack frames and bulkheads at any suspect locations - this is why the temporary self-tappers are so handy. Look carefully at the overlapping edges of planks #1 and #2 to locate any places where they do not sit flush with each other.



The picture above shows a plank being dry-fitted to its neighbour (different boat from Phoenix 15, and built up-side down). See how there are a few self-tapping screws helping to keep the overlapping plank edges snug with each other. I'm conservative and generally put extra temporary screws in to firm everything up – if you look carefully you may see where I have put pencil marks in at even spaces, where more temporary screws will be inserted.

When you are happy that all is ready, remove the temporary fasteners and lay the plank on a bench or the floor, with the inner face upwards. Prime all the gluing surfaces with unthickened epoxy, and working quickly, apply thickened adhesive to one side of all the mating surfaces. Starting in the centre put temporary screws in, moving out on each side of centre. Initially just place enough fasteners in to loosely locate the plank and then go back and methodically drive all the required screws. Check to confirm an even squeeze-out of epoxy from all joints. Clean up excess epoxy and smooth the remaining squeeze-out into concave fillets. The rounded end of an ice-block stick is good for this job. Remember to make sure the upper edge of the plank is also in contact with the bulkheads and frames.

Once more take the opportunity to check for twist, misalignment and straightness of the hull before walking away to allow the epoxy to cure. Don't forget to clean-up the glue drips on the inside of the hull before it becomes hard!

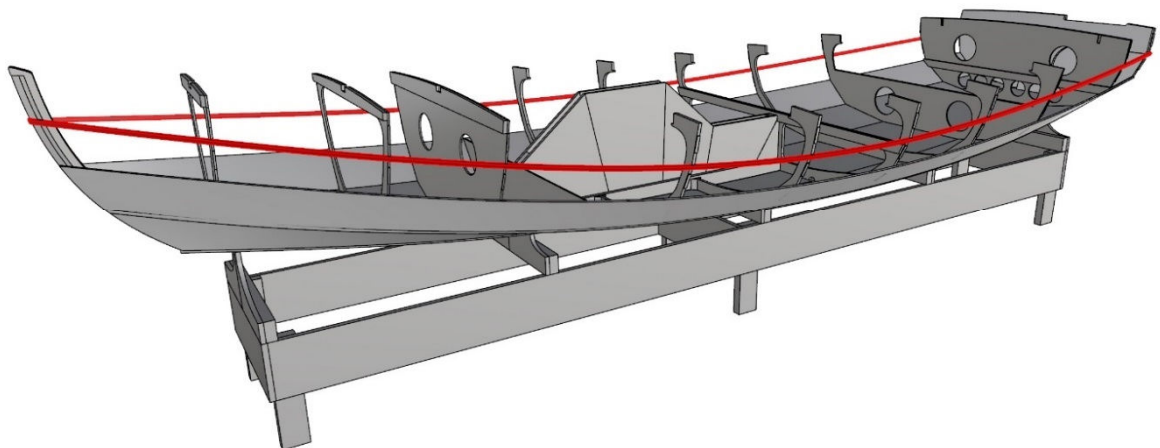
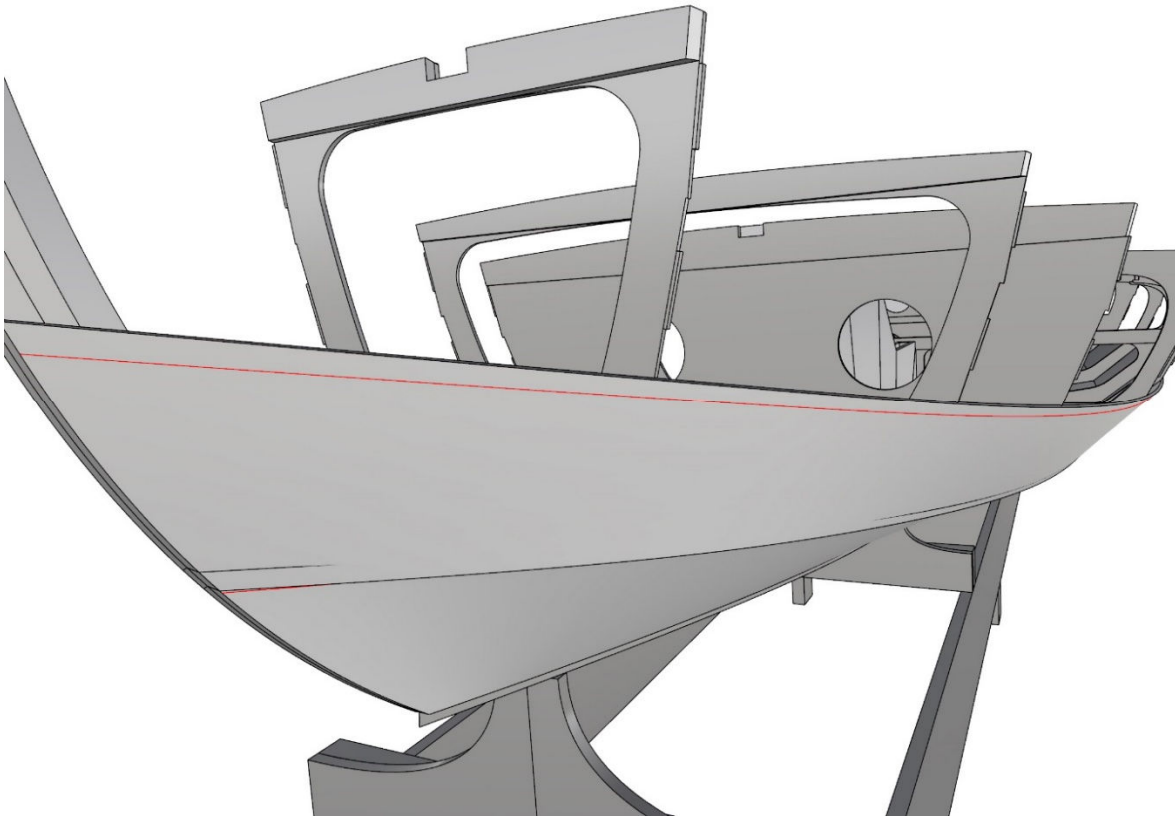


This is how your boat should be looking now. Remember that plank #4 is only temporarily attached to hold the frames, bulkheads and transom in position.

Plank 3

When the epoxy holding plank #2 has cured, it is time to remove the temporary fasteners - remember my advice earlier in the booklet about using a soldering iron to aid removal of screws which have become stuck with epoxy.

Now that the bottom panel, planks #1 and #2, and all the bulkheads and frames have been joined permanently, there is no need for planks #4 to be used as a temporary brace, so remove them and set them aside for later.



Mark a line on plank #2, 15mm (19/32") down from the upper edges as per the above image. This is the bevel line, to be used in the same way as you did on plank #1.

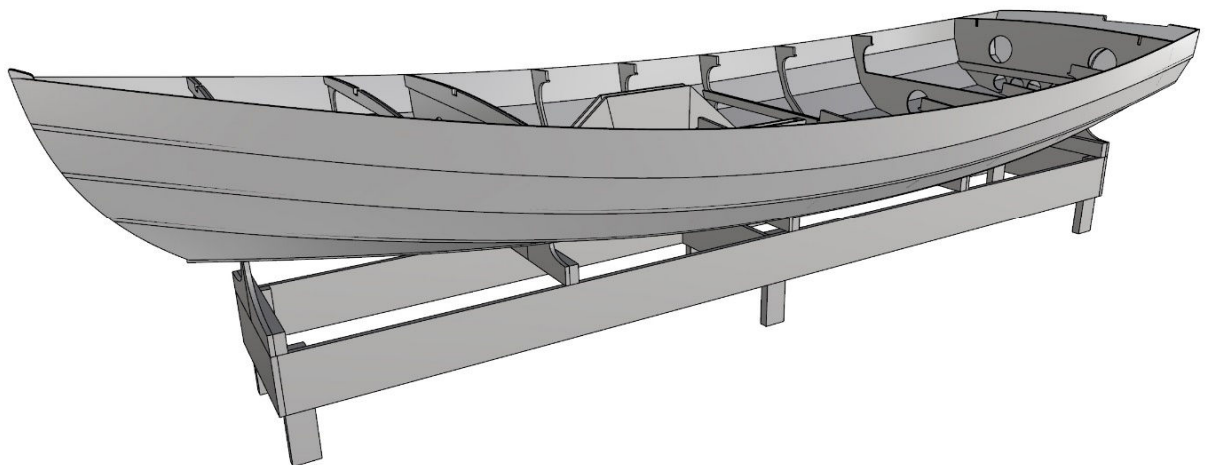
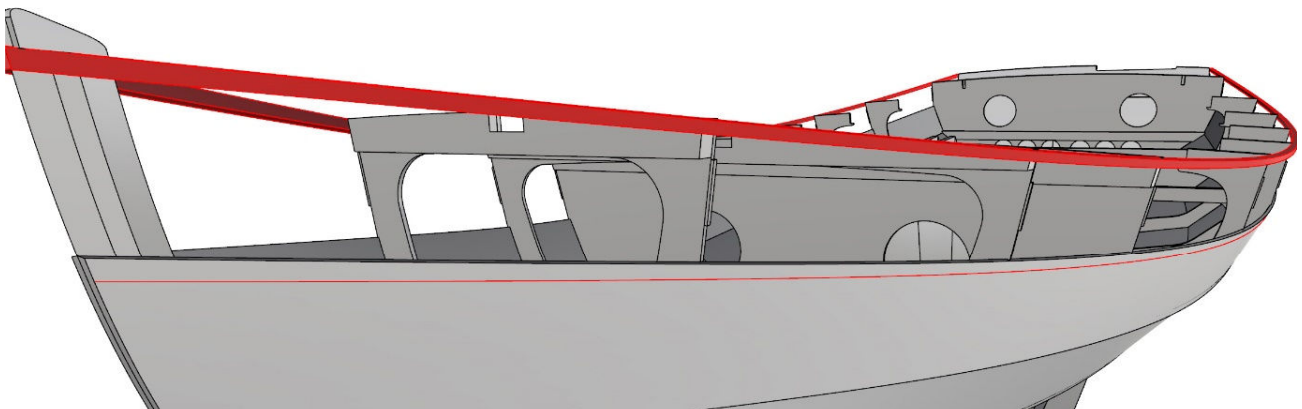
Attach battens to the bulkheads and frames with the top edges touching the notches for planks #3 (see image above). Plane the bevel on planks #2 just as you did on planks #1.

Repeat to process used when dry-fitting and then gluing previously

Now for the last pair of planks. Attach the batten and mark the 15mm (19/32") bevel lines as before.

Repeat to process used when dry-fitting and then gluing previously.

Celebration time - the hull is fully planked!



Next the gunwale...

The gunwale is made from a durable hardwood or tough softwood such as Hoop Pine or hard Douglas Fir. I recommend 38mm x 12mm (1-1/2" x 1/2"), tapering to 22mm x 12mm (7/8" x 1/2") towards the bow and stern – this taper will improve looks dramatically.

After cutting the gunwales to size, dry fit them starting at the stem and working back towards the transom. I generally use spring clamps or small conventional clamps to hold the gunwales in place initially. My strong recommendation is that you install both gunwales at the same time, working progressively aft on one side and the other. The aim is to distribute the bending loads evenly over the hull to prevent asymmetric distortion.

Once the gunwales are positioned correctly using clamps, check the hull by eye to make sure the two sides have fair and symmetrical curvature. Drill for screws to hold the forward ends to the stem – 8ga x31mm (1-1/4") silicon bronze counter sunk head screws are about right for this job. Drive from the outer face of the gunwale into the stem – one screw on each side should be fine. Mark along the inside face of plank #4 for temporary self-tapping screws, spacing them about 150mm (6") apart. (see fig xx). These screws will hold the gunwales in place when you eventually glue them to the hull. Wet epoxy is quite slippery, and using screws instead of clamps when gluing will save a lot of angst.

Continue with the temporary screws until you get to the transom at which point the same screws as used at the stem can be driven from the outer surface of the gunwale into the transom framing.



At this stage you should have both gunwales dry-fitted with temporary screws, and with permanent screws at the bow and stern. See photo below.



Gunwale dry-fitted to a *First Mate*

Remove the gunwales and apply unthickened epoxy to the plywood and gunwale joining faces (faying surfaces) to prime the joints. Then spread thickened adhesive mix to the primed gunwale faces. There is no need to apply epoxy adhesive mix to the plywood because, as long as both surfaces of each joint have been primed, a single layer of adhesive is fine.

Starting at the bow start to reattach the pair of gunwales using the existing screw holes to ensure accurate positioning. The job will be much easier if you have a helper to hold the other ends of the gunwales to prevent them from drooping uncontrollably. However, I have done the job many times on my own, in which case I place wooden planks or long sticks across the boat at 90 degrees to the centreline towards the stern on which the gunwale ends can rest while I insert the screws up at the bow, moving aft from side to side so as to keep each side under similar strain from the bending gunwales.

Cleaning Up the Inside

The boat will never be as clear of obstructions as she is now, so take the opportunity to remove cured epoxy squeeze-out and drips.

Resist the temptation to sand the cured epoxy from the surfaces. Sanding will inevitably result in the surrounding wood surfaces being sanded before the gobs of epoxy are gone, leaving a depression which has to be filled. The best way of removing the epoxy drips is to warm them with a heat gun on a moderate setting, and shave them off flush with the surface of the surrounding timber using a sharp scraper. The technique takes a

little while to master, but once you have the hang of it, the job goes rapidly. When the gobs and drips are removed, a general sanding using a random orbit sander and/or a sanding block will get the interior smoothed up in good time.

Installation of Deck

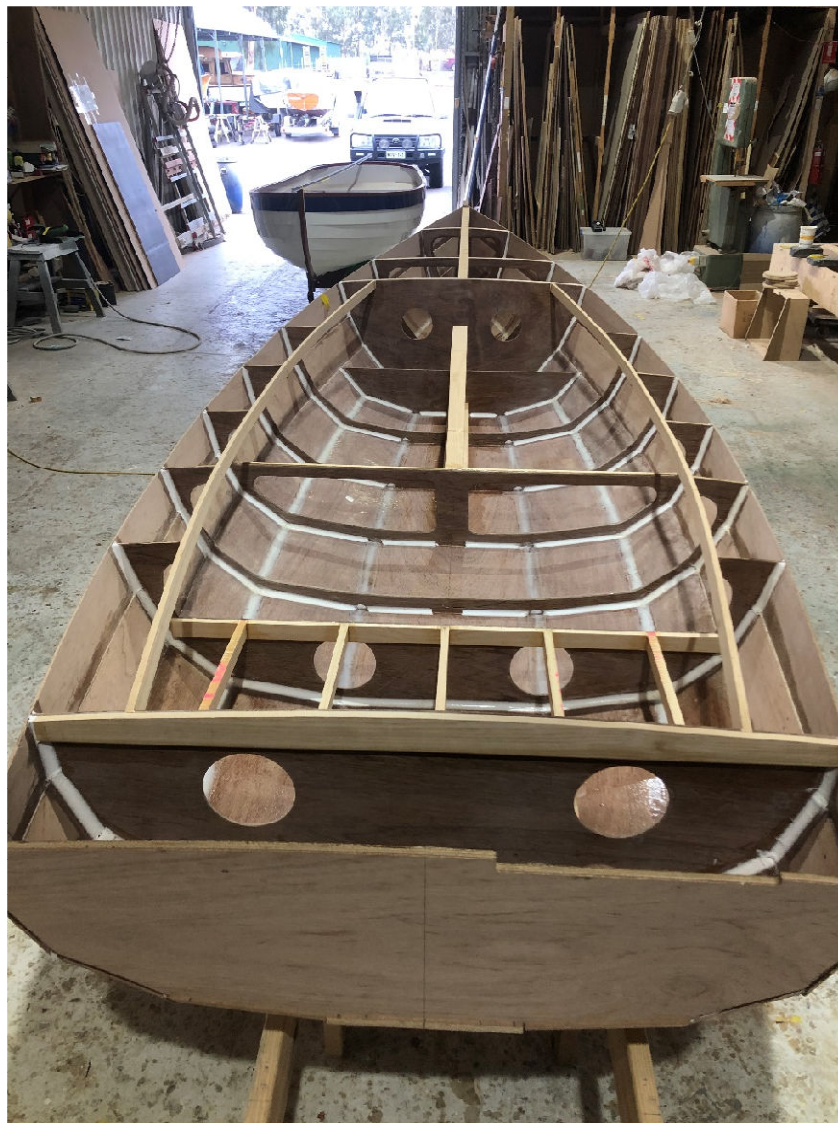
The deckbeams should already be in place, as they are incorporated in the construction of the permanent bulkheads #1, #2, FRD BHD, AFT BHD and Transom. Side deck knees form part of frames #4, #5, #6, #7 and #8.

What remains to be done is the installation of the kingplank, and the aft deck framing (including the self-draining outboard well).

Most details should be evident in the various sheets of the plans. Use your own judgment about exact proportions and dimensions, but the plans represent much careful thought on my part, and all things have been done for a reason. Make major changes at your own peril, and don't blame me if things don't work out further down the track!

A couple of general points –

- The longitudinal deck beams below the aft deck and the stern sheets (i.e. aft seat) do not need to be notched into the bulkheads and transom. You can do it if you like. But if they are properly glued to the underside of the deck when it is installed, the structure will be strong enough without notching;



I am not a fan of coating plywood or natural timber with epoxy except when combined with glass or DYNEL cloth, the reason being that it is too easy for the unreinforced epoxy to be damaged and allow water entry. However, when laying decks, I usually roll at least three coats of epoxy onto what will end up as the underside of the deck. The deck undersides are well protected from the sun's UV radiation and are not subject to damage from fallen objects nor gritty shoes, so I feel the epoxy-only coating will survive long term in that situation.

My method involves doing all the pre-fitting and pre-drilling with the ply in its natural state. Once I'm sure that everything is correctly positioned and dry fastened, I take it all apart and roll at least three epoxy coats onto the undersurfaces, leaving two to four hours between each coat – this is so the epoxy has cured enough so that it won't get lifted off during the laying of the subsequent coats.

While waiting for the last coat to get tacky, I apply a priming coat to the gunwales and top edges of the planking, all the deck beams and top edges of the side deck carlings. Then when I'm ready I apply a bead of thickened epoxy adhesive mix to each section to be covered – this only needs to be applied to the primed joining (i.e.faying) surfaces on the boat because as long as all the joining surfaces are primed, only one side needs the thickened adhesive mix. Go ahead sequentially, putting adhesive on the surfaces which will receive each panel of the deck. Get each panel screwed down using the prepared screw holes and when done, repeat the process with subsequent panels.

Get the entire deck down in one day so that none of the priming epoxy and adhesive mix cures beyond the point where it is chemically active for bonding. If it turns out to be impossible to get the job done in a day and the epoxy cures, wait until the next session and abrade all the gluing surfaces with 80 or 120 grit paper so as to get a good mechanical key for the next gluing session.

Remember, the planking extends to cover the gunwales. See detail drawing in the plans. I find that the joining edges of the plywood deck sections can be butt-joined with a plywood butt-strap underneath. For butt-straps on 6mm (1/4") ply, 100mm (4") wide straps are fine i.e. the butt-strap overlapping the joint by 50mm (2") each side. Photo below shows a place on the deck where a butt joint will occur. The 100mm butt-strap will extend 50mm (2") under the deck already in place and the next section of side deck will sit on top of the remaining 50mm (2") of the butt-strap.

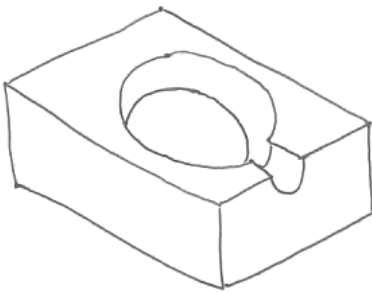


Mast Step and Mast Partner

The information provided on the plans is quite detailed and instructions are printed with the drawings.

Just a couple of points may be needed for clarification: -

- The face of the mast partner which is glued against bulkhead #2 needs to be beveled approximately 5 degrees to allow for the aft rake of the mast;
- Drainage must be provided when making the mast step. This can either be a hole drilled into the side of the mast step at the bottom edge, so that water can trickle out into the bilge of the boat. Another method is to cut a groove into the underside of the step



Above drawing shows the mast step upside down to illustrate the idea of the groove cut into the bottom. When installed the right way up, the groove would form a sort of tunnel with the keelson as the floor of the tunnel.

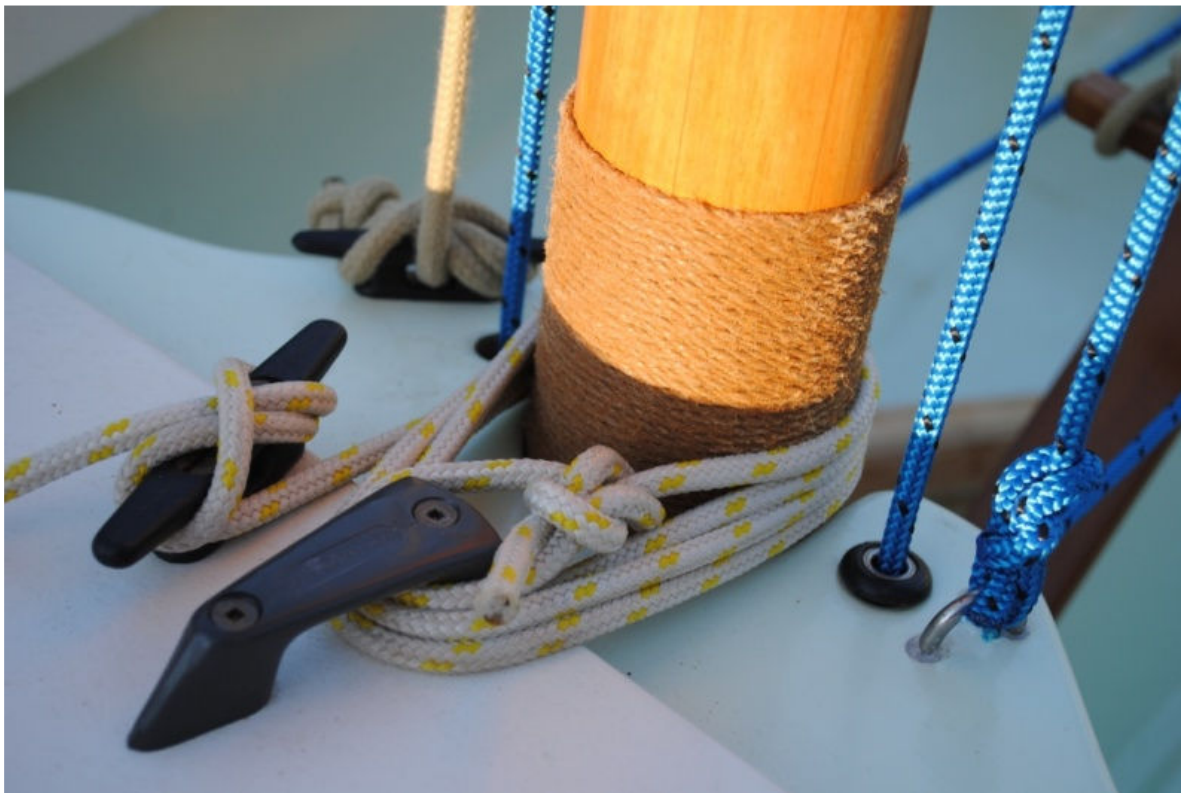
Make sure that the forward face of the mast step is strongly glued to the face of the bulkhead.





The bronze gate in this photo is a bit light, and should be replaced with a heavier unit, but it illustrates the principle.

Alternatively (and my preference), a lashing can be employed to hold the mast in place.



Centreboard Case Details

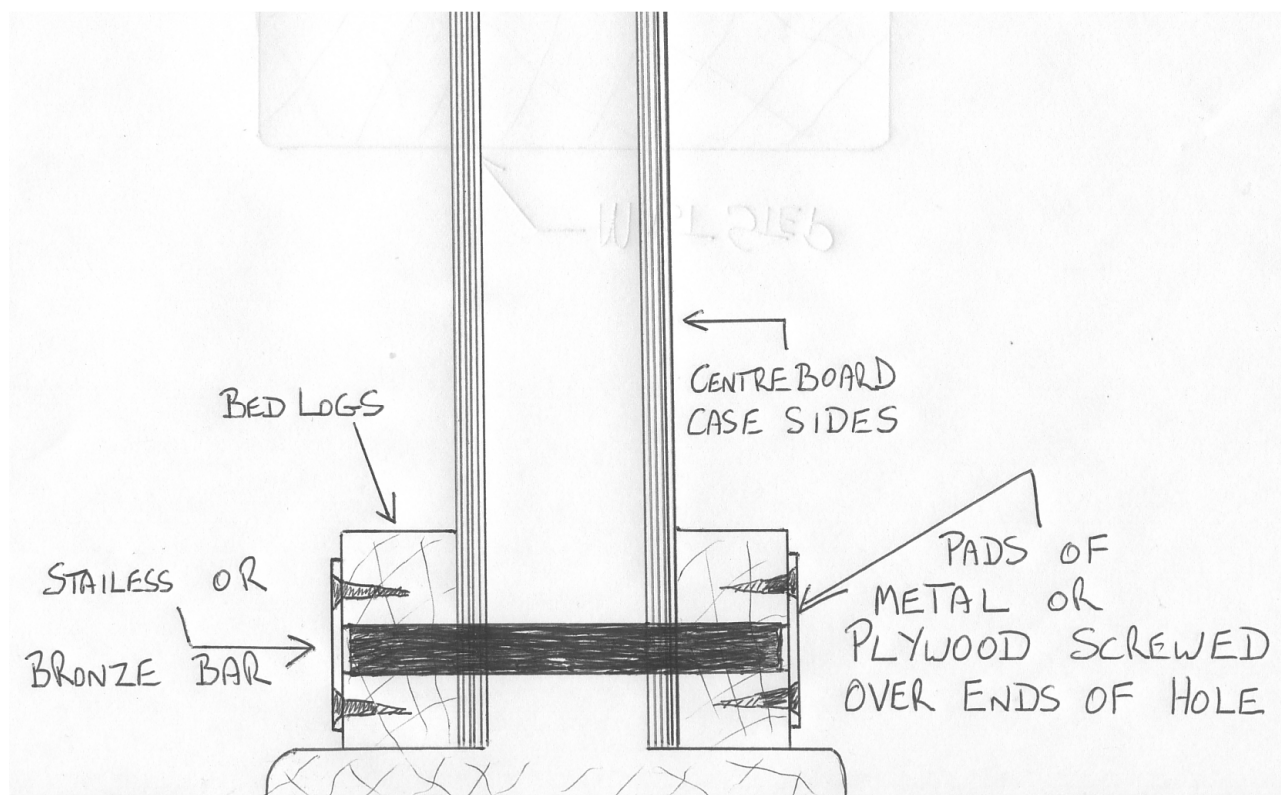


The centreboard raising and lowering rod is well dimensioned on the plans. However, here is photo which illustrates the principle of operation. When the board is raised and the rod is folded forward, it rests on the top of the case cap, and holds the centreboard in the raised position securely. It is good to reinforce the case capping where the rod bears on its surface near the hole. Use bronze, brass or stainless-steel plate. In this photo, Paul Hernes has also placed a pad of leather up near the handle end to protect the paintwork. The rod can be lashed down or clipped under a little thumb-cleat.



The above photo shows a deluxe version made by Byron Bennett, which incorporates intermediate lowering positions.

The pivot bolt can be done several ways, but beware – many systems leak, and cause local stresses. The method I prefer is to pivot the centreboard on a piece of plain bronze, or stainless steel bar which is made fractionally shorter than the overall width of the centreboard case including the bed logs.



In the above drawing you can see how the bar is lying loose in the hole through the centreboard and the bed logs. Over the ends of the hole, place a small plate of bronze, stainless steel, brass, or plywood. Make sure to bed the cover plate in low modulus bedding compound such as 3M 4200 or Sikaflex 291 and hold in place with short screws driven through the four corners of the plate.

Painting

Painting is a whole subject on its own, and involves many options and alternatives. The one thing I can tell you for sure is that there are no quick-and-easy ways of doing it.

There are hundreds of products on the market, some of which promise to take all of the work out of applying the finish and claim to be virtually maintenance-free. Despite many years of boatbuilding, I have yet to find a miracle coating which works better than ordinary paint. Some people like to seal all surfaces with un-thickened epoxy. My experience is that epoxy does not penetrate timber more deeply than a tenth of a millimetre or so, except on the end grain. The shiny surface of the cured epoxy is difficult to fair, and paint does not adhere well to the surface unless it is very well abraded. In fact, many paints will not ever cure properly when applied over some epoxy products.

If you like the idea of using epoxy, go ahead and use it. But please make sure that you follow the instructions to the letter. Remember, epoxy must be coated with paint or varnish to protect it from its number one enemy – U.V. You will need plenty of paint, and if using varnish, it is necessary to apply at least eight or nine coats of a varnish which has the highest possible level of U.V. filtration. In this case, it is also essential that you use an epoxy which has been specially formulated to be applied under a clear coating.

I still use epoxy coatings for some locations such as the internal surfaces of buoyancy tanks and inside hollow masts. In these cases there is little likelihood of UV damage, and the epoxy surface is not going to be stood on by sandy shoes, nor have sharp, heavy objects dropped on it. Just remember that if there is any physical damage caused to an epoxy coated surface, water will get in, but it will have great difficulty getting out.

Think of a dry sponge completely covered with a plastic film and then putting a small cut through the plastic film with a razor. Place the sponge under water in a bucket for a few hours and then pull it out and place it out in the sun. My guess is that the sponge will remain wet for days or weeks, and simply go rotten.

I do use epoxy coatings occasionally on bare timber external surfaces prior to painting with primer, but when I do, I use one of the very much thinned preparations which claim to penetrate deeply. I do not believe that these actually seal the wood, because they have so much volatile thinner in the mixture. However, when using a particularly low-density plywood or timber, I brush a single, light coat on to densify the surface a little, and to raise the grain for initial sanding. Such a light coating of thinned epoxy will allow any water which gets into the timber to migrate back out through itself and the slightly porous paint coatings.

I use single-pack paints for reasons of personal safety from poisons, ease of repair, and their somewhat porous nature.

For what it is worth, our normal painting schedule is as follows: -

- Thoroughly finish sand through a range of grits, stopping at either 120 grit or 180 grit when painting, and going through to 180 grit or 240 grit when varnishing;
- Apply a thinned primer – we usually thin the first coat with the addition of about 40% thinners. Always use the thinners recommended by the paint manufacturer – don't just put in turpentine unless recommended by the manufacturer. In the case of a varnished finish, we thin the first coat 40% or even 50% so that it acts as a penetrating priming coat;
- Apply at least four coats of primer/undercoat, sanding between each coat. Start off with about 180 grit, and move through to 240 grit for the third and fourth fill coats;
- Finally apply at least three coats of top coat, sanding lightly between coats with 240 grit to 320 grit;
- In the case of varnish, apply about eight or nine coats after the initial thinned priming coat, sanding between coats with 240 grit to 320 grit for dry paper, or 400 grit for wet sanding.

All of the above may sound like hard and tedious work, but in the long run, it is the best way. In future years, you will appreciate the good foundation which you laid down.

Best of all is for you to research the subject of coating for yourself, and follow the paint manufacturer's instructions.

Recently I've been experimenting with the use of water-based paints, particularly above the waterline in areas which are subject to large amounts of UV radiation. Many highly regarded authorities have been pushing the use of water-based paints for years (e.g. Harold Payson, Phil Bolger, Dave Carnell, and, if I understood correctly, John Gardner).

These paints are what are called 'Latex' in the USA, but they don't contain latex.

The ones to use are 100% Acrylic, water-based house paints of the very highest quality available. Examples (in Australia at least) include Dulux Weathershield, Wattyl Solargard, and Taubmans All Weather Exterior.

I'm still a long way from being totally convinced about these paints, but things are looking very promising for trailer boats.

My current approach is to use a high-quality primer as described earlier for oil-based paints. I'm experimenting with both oil-based and water-based primer/undercoats, but haven't yet made up my mind which is best.

For application by brush, a foam brush seems to be the best, given that the thick water-based paint is prone to brush-marks. For the best finish you really need a professional-standard airless spray unit, but

these cost many thousands of dollars, and are potentially extremely dangerous due to the risk of paint being injected through the skin of a careless operator or helper. I bought mine second-hand for \$1,200.00, but expect to pay nearly \$4,000.00 for a decent new unit.

Probably the most practical compromise is to use a roller and brush for a 'Roll-and-Tip' application.

I continue to use Boatcraft Pacific 'Aquacote' water-based epoxy primer/undercoat, and 'Aquacote' water-reducible two-pack linear polyurethane for serious work. It is very good stuff indeed, but you MUST follow the instructions to the letter in order to ensure success. The manufacturer (Boatcraft Pacific) has invest a huge amount of time and money into the development of this paint system, and they know the best way to get results. Follow their instructions.

'Aquacote' is an Australian product, but if you are in a location where it is not available, there are a limited number of similar products on the market.

Once cured, the resulting paint film is very tough, and just as durable as the solvent-based two-packs.

Rudder and Tiller

Rudder details are shown in a good level of detail in the plans. However, here are a few points which may help:-

- Apply 200gsm/6oz glass cloth to the inside faces of the two cheeks of the rudder box. This is to protect against abrasion as the rudder blade is raised and lowered. Also, apply glass to each face of the filler piece. This is necessary so that the filler piece ends up at least as thick as the rudder blade;
- After fairing to a hydrofoil section, cover both faces of the rudder blade with 200gsm/6oz glass cloth set in epoxy. The safest method is to apply the glass to one face – allow it to cure, and then trim the overhanging cloth. Turn the blade over, and apply glass to the second face. Do not attempt to run the glass around the leading and trailing edge – it won't take the sharp bend;



Cover one side of centreboard or rudder with 200gsm/6oz woven glass and epoxy...



...then after the glass is cured enough to be stiff, turn over the foil and squeeze in a good bead of thickened high-density epoxy...



...and fair the cured filler to a good shape on the side which hasn't been 'glassed...



...and then glass that side, allowing the glass cloth to simply run out over the faired edge. When the glass/epoxy has cured, you can trim the cloth and carry out final sanding of the edges. The glass will comply with the surfaces of the foils better, and there is a tough, abrasion-resistant covering to protect the plywood leading and trailing edges.

Purchase or make rudder gudgeons and pintles of your choice. I prefer cast bronze fittings, but they are difficult to obtain in the correct sizes. Good quality marine stainless-steel units are readily available. Check that the gap between the tangs is correct for the thickness of the completed rudder box. Also beware of driving fastenings through the cheeks at bottom end of the rudder box and into the rudder blade!

Rigging

The recommended rig for *Phoenix 15* is the 76 sq.ft balance lug and I say this for a number of reasons:-

- Simplicity – the mast is free standing, without any stays, the sail remains attached to the yard and the boom, and there is no attachment of the sail to the mast. All of this means quick rigging and de-rigging at the boat ramp;
- Short mast, yard and boom – rig doesn't overhang anywhere when towing;
- The sail is particularly easy to reef, and when reefed, sets efficiently;
- By hauling the forward end of the boom aft to the mast, the sail converts to a standing lug configuration and a jib can be set flying changing the boat into a sloop. The process can be reversed in a few minutes turning her back into a balance lug.
- A properly set balance lug is a powerful rig for the length of the mast and other spars, requiring few store-bought fittings.

Mast, Sprit and optional Boom

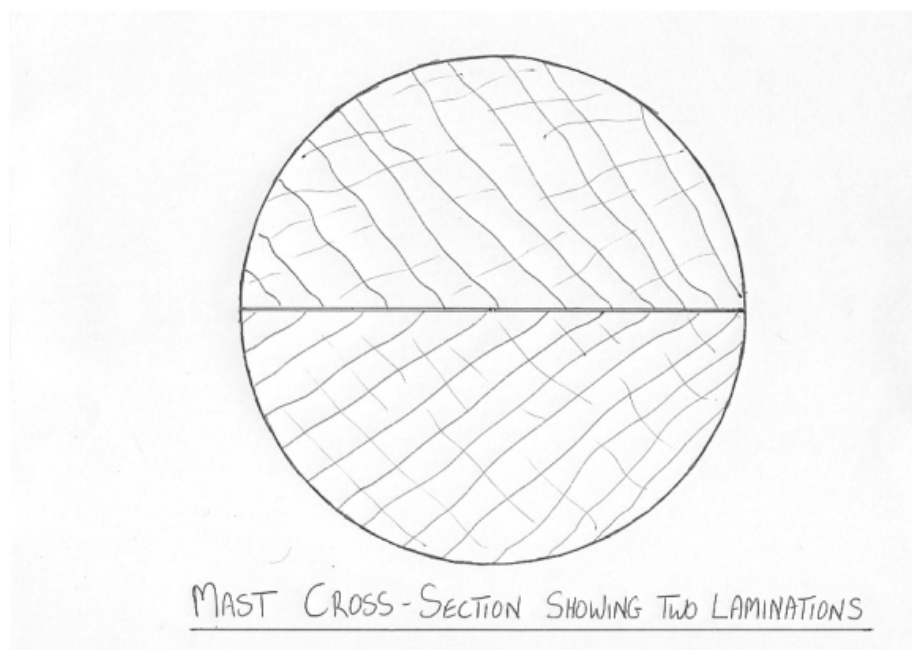


See mast above – (this is a 'Bird's Mouth' mast in the process of being planed down to a round finish).

The size specified for the mast in the plans drawing will provide adequate strength even if the mast is built hollow

The decision to go for a solid mast or a hollow mast is entirely up to you. I usually make mine hollow, because I've found that it is no more labour intensive, and the resulting mast is significantly lighter. However, the mast for *Phoenix III* is very short relative to the sail area it supports, so the weight penalty of using a solid mast is not too bad.

Even if you have a solid stick of timber which is a suitable length, breadth and thickness to make the mast in one piece, it is advisable to cut the stick into two pieces, turn one half end-for-end, and then glue the halves back together. The resulting grain pattern will be symmetrically opposed, something like the example shown in the drawing below.



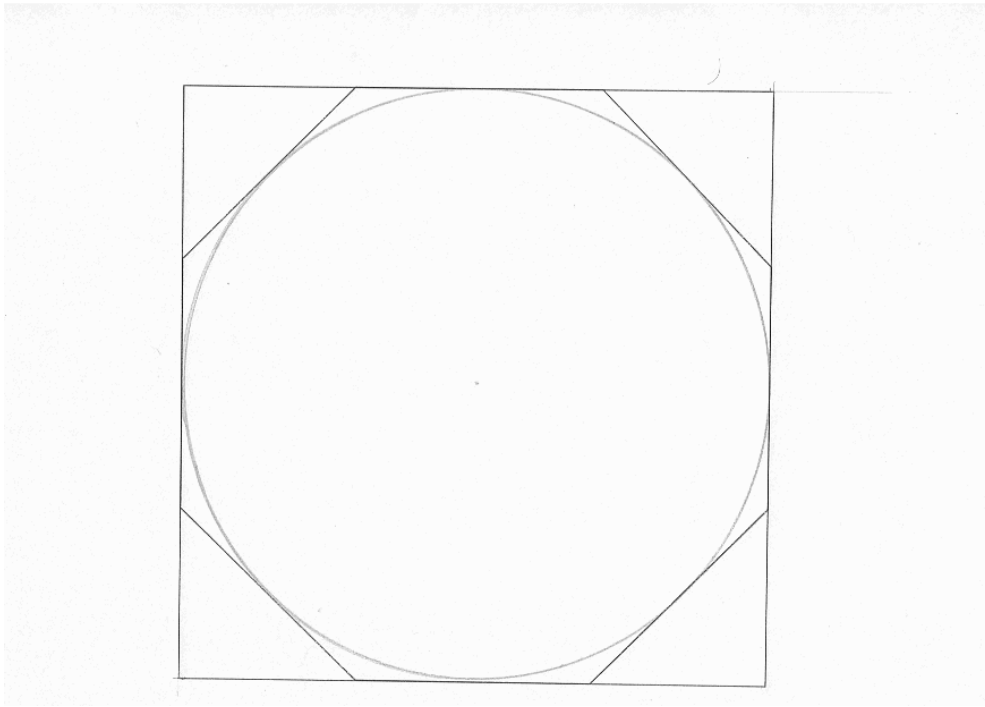
The reason for taking the trouble to laminate with grain opposed is that the resulting spar will be much more resistant to warping. I recommend the same process be used when making the boom and yard.

When making a solid spar, start off with a laminated section which is at least as wide and thick as the maximum diameter listed in the plans for that particular spar.

It is almost impossible to get the laminated blank perfectly straight, so it is necessary to mark a centreline using a chalkline. After snapping the chalkline, draw section lines (at 90° to the axis of the spar) at the spacing noted on the plans. Mark the section lines with the dimensions listed on the plans. The plan measurements are the diameter of the finished spar, so you need to mark half of that distance either side of the centreline.

Using a stiff batten, draw a fair curve through all of the points. This will give you a plan view of the tapered spar. Using a bandsaw, and/or a plane, cut the taper as drawn. On completion, turn the spar through 90° and repeat the whole process. This should give you a square section spar, tapered as per the plans drawing.

Next, the spar must be marked with lines which will allow it to be planed down to an octagonal shape (see below).



A description of how to produce the tapered lines for the octagonal shape is outside the scope of the instructions. However, many good texts on boatbuilding give an excellent idea of the process. Both of the books I recommended at the beginning of this manual provide all of the detail you will require. These books are, *"Clinker Plywood Boat Building Manual"* by Iain Oughtred, and *"How To Build Glued-Lapstrake Wooden Boats"* by John Brooks and Ruth Ann Hill.

Use a plane or spokeshave to cut the spar to the octagonal shape. It may sound like a big job but you will be surprised just how quickly it can be done.

The next job is to plane off the points of the octagon. For this job, I mark the spar with bands of pencil marks which help me gauge how much material to remove. Place these bands of pencil at about 600mm/2ft intervals. As you plane off the corners, a gap appears in the pencil mark which gives a clear visual indication of how much material has been removed, and how evenly the job is progressing.





After this sixteen-siding operation, redraw the pencil bands to join up the gaps. Then plane the corners off one more time to produce a 32-sided section as shown below.



Here is the same mast at the 32-sided stage.

The final job is to sand the spar to make it smoothly round.

What I have described may sound like a lot of work, but you will be surprised at how quickly it can be completed, and the satisfaction gained from a good job is wonderful. You will have turned a rough lump of timber into a smooth, round, tapered piece of functional art.

For those who would like to have a go at making a hollow mast, I recommend the “Bird’s Mouth” system. This process involves cutting eight tapered staves (or planks if you like) which are glued together to produce a hollow octagonal mast. From that stage on, the process of sixteen-siding, thirty-two-siding, and rounding is identical to that used with the solid spars.

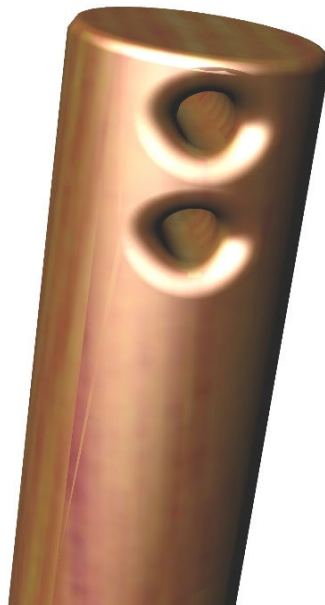
The “Bird’s Mouth” system is covered in various text books, and in excellent articles which have appeared in *WoodenBoat Magazine* and in Chuck Leinweber’s E-Magazine, *Duckworks*. If you do your homework, you will be able to find plenty of information on the process. You can contact me by mail or email for more help.

Tips of Mast, Yard and Boom

My main aim has been to keep the rigging arrangements as simple and reliable as possible – this also means that you don’t have to spend money on expensive fittings.

The following relates to the lug rig. The arrangement I describe is just one of a large number of options. You can make up all sorts of different combinations of blocks, cleats and lines - read some books, and use commonsense. But remember, most changes will end up costing you money on factory-built marine fittings. My system is simple, cheap and reliable – and there is nothing to stop you from modifying and developing as you gain experience with the boat.

The mast head has two holes, both 12mm (1/2”) diameter. Drill both at 90° to the centreline of the boat. The top one takes the lashing which holds a pulley block on the forward face of the mast to carry the jib halyard.

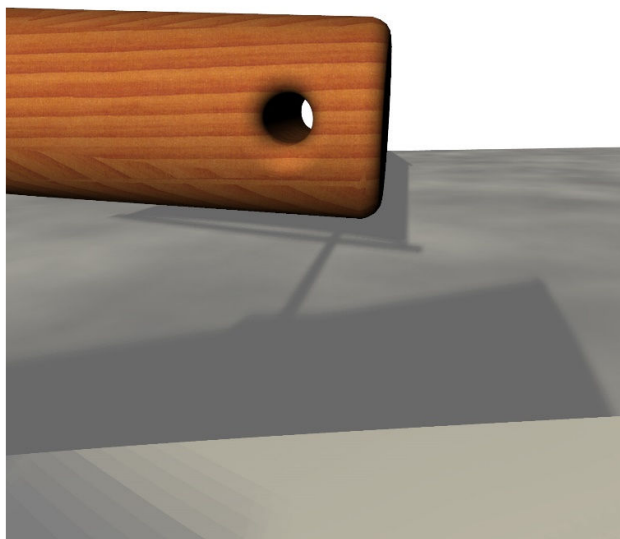


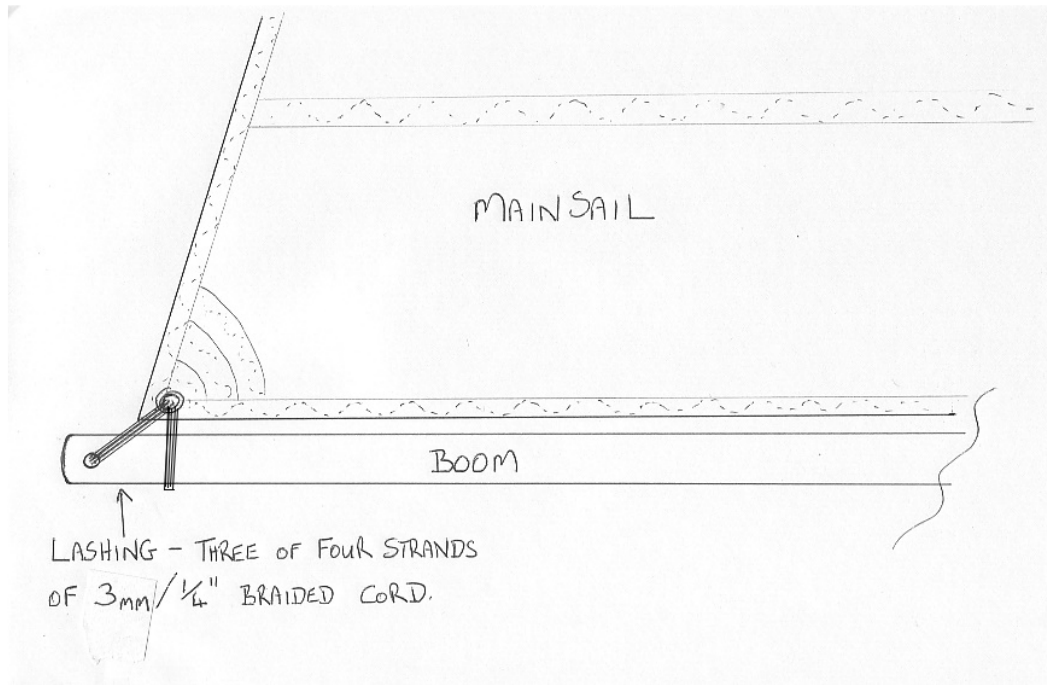
The second hole carries the mainsail halyard. Round off the entry and exit points to reduce friction (see drawing above for an illustration of the idea).

Both the main and Jib Halyards are made off on cleats or belaying pins on the mast partner.



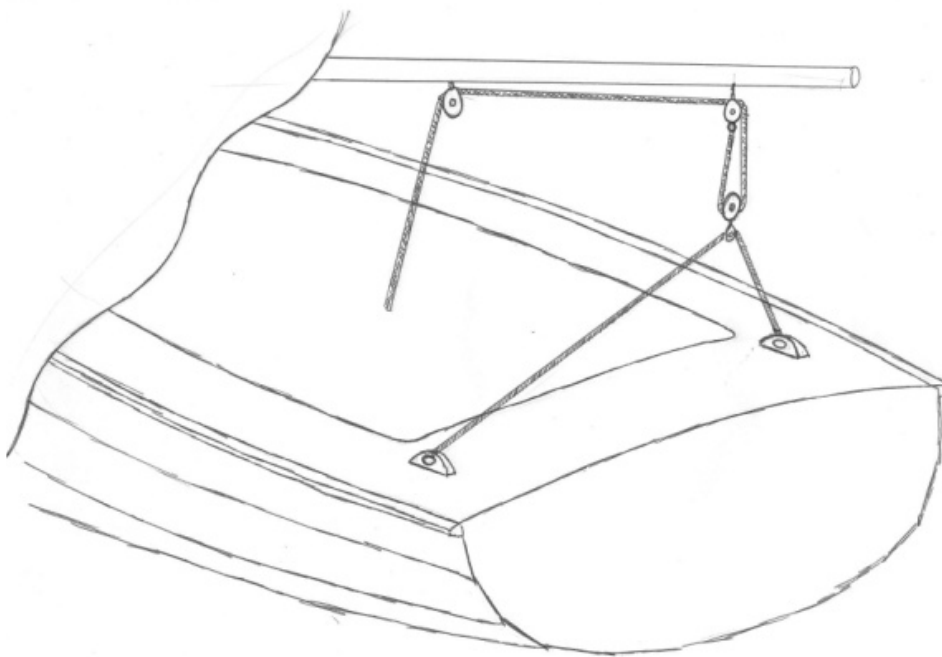
The boom and the yard both have 9mm (3/8") holes at each end to take peak and throat, outhaul and tack lashings made up of about four runs of 3mm (1/8") braided polyester venetian blind cord.





The sketch above shows the general idea of how the four corners of the balance lug mainsail is attached to the ends of the boom and the yard.

Main Sheet and Jib Sheet Fairleads



This shows just one of many ways to set up the mainsheet. The block on the bridle is free to run from side to side, and it helps to have a small block attached to the main block to reduce friction as it runs from side to side. There are many other arrangements possible, but this one keeps clutter out of the inside of the boat. I use about 9mm (3/8") diameter braided Dacron line as a main sheet and as the jib sheets.

Jib sheet fairleads should be mounted on the deck on the inner edge of the side decks – i.e. mounted with

bolts or screws driven vertically through the deck and into the side deck carlings. This location should be measured from the forward face of the stem back to the side deck inner edge. The distance should be 1685mm (5' 6- 5/16") as a starting point. Move the fairleads forward or back if required to get the jib setting to your preference. See the plans for a visual explanation.

The tack (lower, forward corner) of the sail must have a downhaul of some type in order to keep tension in the luff of the sail.



In the preceding photos you can see how the downhaul is arranged in Paul Hernes' boat. He has used a simple lashing down to a stainless-steel lined fairlead.

As with all rigging set-ups, there are many different arrangements which you can use, and part of the fun involves making modifications as you gain experience. An easily adjustable downhaul is useful for sail shape adjustment – systems can easily be devised using pulleys and jamb cleats.



This is a system I installed on a 'First Mate' I built for a friend. Both the halyard and the downhaul run through holes on each side of the mast partner and then through fairleads and back to V-Jammers on either side of the centerboard case. The idea was to make it easy for a solo skipper to make adjustments without moving forward. The above photo shows the halyard on the starboard side.



Here you can see the downhaul on the port side. The rigs for 'First Mate' are identical to those for 'Phoenix 15' and 'Phoenix III'.



In this photo you can see my boom parrel arrangement. The aft end is just a bowline with a loose loop around the boom just in front of the downhaul. From there it runs around the mast and then forward to a fairlead at the forward end of the boom.....



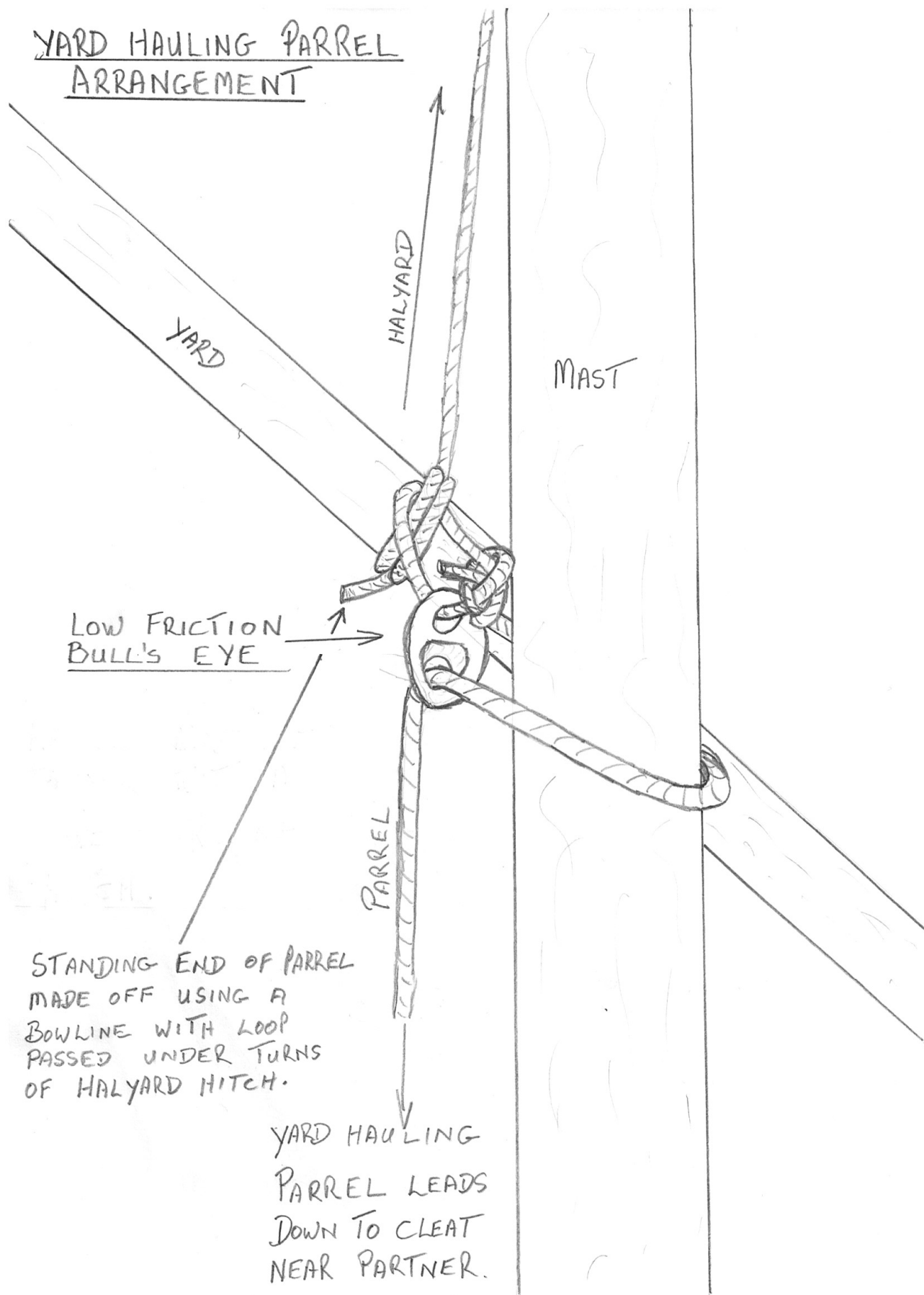
...and then back to a horn cleat on the underside of the boom to be within easy reach from within the cockpit. This system holds the boom securely against the mast, and by releasing the line from the horn cleat you can haul in on the line to bring the forward end of the boom back to any location you wish, including all the way back to change the mainsail configuration from a balance lug into a standing lug. This is particularly useful if you decide to set the optional jib set flying.



The halyard (the blue line shown in this photo) runs through an athwartship (i.e. 90 degrees to the centerline of the boat) hole with faired edges and down to be tied to the yard using a rolling hitch. The buff-coloured line you can see is a 'running parrel' or 'yard hauling parrel' which is optional and will be explained on the following pages.

The head of the sail is attached to the yard using individual ties, usually called robands. For these I generally use 3mm (1/8") venetian blind cord or similar.

YARD HAULING PARREL ARRANGEMENT



Oars and Oarlocks

Phoenix 15 has been designed from the outset to be a pleasure to row.

Most people go through life without ever discovering how satisfying it is to row using the correct equipment. Of critical importance is: -

- Correct location of the oarlocks in relation to the rowing thwart;
- Correct length of oars;
- Correct relationship between the length of the oar inboard and outboard of the oarlock;
- Correct handle shape;
- Correct oar balance.

The oarlocks should be positioned approximately 180mm (7") above the level of the rowing thwart, and 300mm (12") aft of the aft edge of the thwart (see page 24-A3 of the plans).

When choosing oarlocks and oarlock mounting plates, consider whether the oars will clear the inner and outer edges of the side decks when in use. It may be necessary to place the mounting plates on blocks to raise the oarlock slightly.



In the plans, I have included full-sized drawings for oarlock blocks.



Above is a selection of different oarlock and oarlock mounting plate styles. For *Phoenix 15* I would choose the second from the left as being the most appropriate.

If you have trouble locating a supplier for such items, contact me and I will give you advice. (I keep oarlocks in stock at most times).

For *Phoenix 15*, I recommend 2300mm (7-1/2') oars as being the most suitable. The rough rule-of-thumb for oar proportion is 68% of the length outboard of the lock and 32% inboard.

Typically, oars sold on the retail market are made on a lathe and are very badly balanced. The blades are usually much too thick and heavy, and there is not enough weight in the oar towards the handle.

Custom made oars are very expensive, but you can make your own, and it is a satisfying (but demanding) process. An easier approach is to purchase standard 7-1/2 footers, and then plane down the blades to make the weight distribution better.

Suitable patterns for oar making are available from Mystic Seaport Maritime Museum, WoodenBoat Magazine and other such sources. Once again, if you are really stuck, contact me for extra detail.

Leathering

Oars and the sprit will benefit from leather chafe protection. Details on leathering can be found in many nautical books, and once again, from those wonderful resources – WoodenBoat Magazine www.woodenboatstore.com and Duckworks Magazine www.duckworksmagazine.com

Here is me struggling with a leathering project assisted by my son, David.



And here is the result



The sprit and/or the mast can be similarly protected

Recommended Reading

- **WoodenBoat Magazine**
- **Duckworks Magazine** www.duckworksmagazine.com
- **The Gougeon Brothers on Boat Construction** by Meade Gougeon
- **Clinker Plywood Boatbuilding Manual** by Iain Oughtred
- **Boatbuilding for Beginners (and beyond)** by Jim Michalak
- **How To Build Glued Lapstrake Wooden Boats** by John Brooks and Ruth Ann Hill
- **Instant Boats** by Harold Payson
- **Build the New Instant Boats** by Harold Payson
- **Boatbuilding with Plywood** by Glen L. Witt
- **The Sailmaker's Apprentice** by Emiliano Marino
- **Brightwork** by Rebecca Wittman
- **The Brightwork Companion** by Rebecca Wittman
- **The Compleat Cruiser** by L. Francis Herreshoff
- **The Common Sense of Yacht Design** by L. Francis Herreshoff
- **The Nature of Boats** by Dave Gerr
- **Boat Strength** by Dave Gerr
- **Boatbuilding in Your own Backyard** by Sam Rabl
- **From My Old Boat Shop** by Weston Farmer
- **Of Yachts and Men** by William Atkin
- **103 Sailing Rigs** by Phil Bolger
- **Small Boats** by Phil Bolger
- **The Folding Schooner** by Phil Bolger
- **Different Boats** by Phil Bolger
- **30-Odd Boats** by Phil Bolger
- **Boats with an Open Mind** by Phil Bolger
- **Schorpioen** by Phil Bolger
- **Sail and Oar** by John Leather
- **Open Boat Cruising** by John Glasspool
- **The Dory Book** by John Gardener
- **Building Classic Small Craft** by John Gardener
- **Canoe Rig – The Essence and the Art** by Todd E. Bradshaw
- **Boats, Oars and Rowing** by Pete Culler
- **Skiffs and Schooners** by Pete Culler
- **Ocean Crossing Wayfarer** by Frank and Margaret Dye

...plus many, many more...

Materials List

- 6mm (1/4") Marine Plywood - 6 sheets;
- 12mm (1/2") Marine Plywood – 3 sheets
- 65mm x 19mm (2 9/16" x 3/4") Hoop Pine, Meranti, Douglas Fir etc
9.5 metres (32 ft)
- 42mm x 19mm (1 5/8" x 3/4") Hoop Pine, Meranti, Douglas Fir etc
12 metres (40 ft)
- 4.8 lt (1 gallon) West System® Brand Epoxy (or equivalent) plus glue/filleting additive to suit
- Mast, sprit and boom timber not included due to construction options.

Please note: - this is only a partial list, and is an estimate only. Take time to study the plans and make a list of what you think may be required. This is a very good way of establishing an understanding of the construction process.

Rigging and Fittings

The selection of fittings and rigging is difficult to specify, as there are a number of rig options on the plans, and the selection of components is very much a matter of personal taste. Commercial fittings are expensive, and the range is wide – from the most basic, to the most exotic. Many people choose to make some of their own fittings and rigging components, and given the simple nature of the rigs, this is a great idea.

Here are some part numbers for components which are suitable (I am referring to the Sprit-Sloop rig with the optional boom). I have chosen from the basic ranges in each manufacturer's line-up. For most people this will be more than adequate, but if in doubt (or if you have lots of money!) go and browse through a major chandlery for alternatives. My earlier page of recommended reading will give you plenty of ideas if you want to improvise for yourself. Start with the most simple gear, and alter as you gain experience. I strongly recommend that you make as many fittings as possible using your own hands and resources.

Rudder Gudgeons and Pintles

- Racelite RL-390 G (1-3/8") Complete Set; or
- Ronstan RF2503 x 2 and RF2515A

Jib Sheet Fairleads

- Racelite RL-850 x 2; or
- Ronstan RF59 x 2 (if fairleads with cam-cleats are preferred, RF58 x 2)

Mainsheet Fairleads

- Ronstan PNP-124 x 2; or
- Ronstan RF59 x 2 (if fairleads with cam-cleats are preferred, RF58 x 2)
- If V-Jamb Cleats are required, use Ronstan RF5110

Snotter Tackle

- Racelite RL-301-B x 1 and RL-302-B x 1; or
- Ronstan RF188 x 1 and RF185 x 1

Turning Block for Snotter Line

- Racelite RL-306-B; or
- Ronstan RF917 Upright Lead Block

Jib Halyard Block

- Racelite RL-301-B; or
- Ronstan RF174

Mainsheet Blocks

- Racelite RL-710-C x 2 and RL-702-C x 1; or
- Ronstan RF51100 x 2 and RF51110 x 1
- (if a ratchet block is desired, the best value is Riley RM324)

Halyards

- Main Halyard and Down Haul from 6mm Braided Dacron. (6mm Spectra can be used, but is much more expensive. It is less stretchy but these components are not highly stressed). You can use 8mm Braided Dacron if preferred.
- Jib Halyard from 6mm Spectra

Snotter

- 6mm Spectra. This component is subject to substantial loads, and will have a bad effect on performance if it stretches. Spectra is worth the cost.

Main and Jib Sheets

- 8mm or 10mm Braided Dacron

General Lacing

- 3mm VB (Venetian blind) cord. Available from chandleries in a range of colours.